



UNIVERSITY OF PATRAS
DEPARTMENT OF ELECTRICAL AND COMPUTER
ENGINEERING

DIVISION OF POWER SYSTEMS
LABORATORY OF POWER SYSTEMS, RENEWABLE
AND DISTRIBUTED GENERATION

ELECTRICITY PRICE FORECASTING THROUGH MACHINE LEARNING TECHNIQUES

DIPLOMA THESIS

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PATRAS – AUGUST, 2025

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ELECTRICITY PRICE FORECASTING THROUGH MACHINE LEARNING TECHNIQUES

of the Department of Electrical and Computer Engineering student

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was presented publicly at the Department of Electrical and Computer
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29/08/2025

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ΠΑΤΡΑ – ΑΥΓΟΥΣΤΟΣ 2025

Πανεπιστήμιο Πατρών, Τμήμα Ηλεκτρολόγων Μηχανικών και Τεχνολογίας Υπολογιστών.

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Επαμεινώνδας Μητρονίκας
Αναπληρωτής Καθηγητής

Acknowledgments

First and foremost, I extend my deepest gratitude to my thesis supervisor, Dr. Dimitrios Papadaskalopoulos, for granting me the opportunity to explore such an engaging research topic and the overall collaboration throughout the process of this work. He was instrumental in steering me towards this field and specialization, helping me find a research direction that resonated with my interests.

I would also like to thank Konstantinos Plakas for his generous technical assistance and readiness to help and discuss whenever new challenges arose. His guidance and expertise were instrumental.

Special thanks go to my friends, whom I have been fortunate to meet during my studies. Their unwavering support has made these past years productive, memorable, and genuinely enjoyable. The adventures we shared are memories I will cherish for the rest of my journey.

Finally, I owe my heartfelt appreciation to my parents and my brother for their unconditional love, steadfast belief in me, and constant support through both the easy and difficult times. Their encouragement has been my anchor and motivation throughout this endeavor.

ABSTRACT

ELECTRICITY PRICE FORECASTING THROUGH MACHINE LEARNING TECHNIQUES

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The accelerating penetration of variable-renewables, the launch of the European Target Model following the deregulation of electricity markets, and the unprecedented price turbulence of 2022-2024 have made short-term electricity price forecasting a strategic task for market participants and market operators. This diploma thesis presents a head-to-head comparison of four representative forecasting methods, seasonal autoregressive models (SARIMA/SARIMAX), ensemble decision trees (Random Forest), gradient-boosted trees (XGBoost) and recurrent neural networks (LSTM), on the Greek Day-Ahead Market (DAM). A cleaned and gap-filled hourly price series spanning January 2018 to December 2024 is constructed, enriched with publicly available day-ahead system load and specific RES generation forecasts, along with calendar dummies. Each algorithm is embedded in a transparent, rolling-origin evaluation framework that mimics the operational forecasting workflow. All models produce 24-hour horizon forecasts and are assessed with MAE, RMSE, sMAPE, and nMAE. Results show that XGBoost reduces the MAE of a naïve weekly-persistence benchmark by 43% and the RMSE by 33%, establishing this the best performing method. Random Forest follows closely, while carefully tuned SARIMAX remains ~30% less accurate and a single -layer LSTM fails to clearly outperform the benchmark. The analysis highlights the selected feature importance as they account for over 50% of model gain. Error decomposition reveals that more than 90% of absolute errors >150€/MWh occur during evening scarcity episodes, where all models capture timing but underestimate spike amplitude.

Keywords: Electricity Price Forecasting, Day-Ahead Market, Greek power system, Machine learning, Deep learning, XGBoost, Random Forest, LSTM, SARIMA, Renewable energy integration.

ΕΚΤΕΤΑΜΕΝΗ ΕΛΛΗΝΙΚΗ ΠΕΡΙΛΗΨΗ

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Εισαγωγή

Η επιταχυνόμενη ενεργειακή μετάβαση που δρομολογείται σε ευρωπαϊκό επίπεδο έχει στόχο την απανθρακοποίηση του ενεργειακού μίγματος, την αποκέντρωση της παραγωγής και την απελευθέρωση των αγορών. Αυτή η μετατόπιση οφείλεται στις αυξανόμενες ανησυχίες για την κλιματική αλλαγή και στους φιλόδοξους διεθνείς στόχους μείωσης των εκπομπών, παράλληλα με την ταχεία πρόοδο στις τεχνολογίες ανανεώσιμων πηγών ενέργειας, αποθήκευσης και τα εξελισσόμενα πρότυπα κατανάλωσης ενέργειας που δημιουργούν αυξημένες ανάγκες εφεδρείας και ευελιξίας. Παράλληλα, από το 2019 η ελληνική αγορά ηλεκτρικής ενέργειας διέρχεται φάση πλήρους απελευθέρωσης και ενοποίησης με τα υπόλοιπα κράτη-μέλη μέσω της υλοποίησης του Ευρωπαϊκού Μοντέλο Στόχου. Η υλοποίηση των τεσσάρων χονδρεμπορικών αγορών, Αγορά Επόμενης Ημέρας (DAM), Ενδοημερήσια, Εξισορρόπησης και Προθεσμιακή, συνοδεύεται από την σύζευξη με τα ευρωπαϊκά Single Intraday Coupling (SIDC) και Single Day-ahead Coupling (SDAC), επιτρέποντας διασυνοριακό εμπορικό προγραμματισμό και αυξάνοντας τον ανταγωνισμό. Το νέο πλαίσιο διαφανών δημοπρασιών και συνεχούς διαπραγματεύσεως ενέτεινε την τιμολογιακή μεταβλητότητα, όπως φάνηκε δραματικά την περίοδο της ενεργειακής κρίσης 2021-2023, όταν οι τιμές της DAM άγγιξαν ιστορικά μέγιστα.

Σε αυτό το περιβάλλον, η ακριβής βραχυπρόθεσμη πρόβλεψη τιμών καθίσταται ζωτικής σημασίας καθώς επιτρέπει σε παραγωγούς/προμηθευτές να βελτιστοποιούν στρατηγικές προσφορών, σε λειτουργούς συστημάτων να προϋπολογίζουν ανάγκες εφεδρείας και σε χρηματοπιστωτικά έργα να αποτιμούν έργα ΑΠΕ με ρεαλιστικούς κινδύνους τιμολόγησης.

Η παρούσα διπλωματική εργασία διεξάγει μια ολοκληρωμένη, αναπαραγωγική και ποσοτικά αυστηρή σύγκριση τεσσάρων ετερογενών παραδειγμάτων πρόβλεψης ωριαίων τιμών DAM κατά την περίοδο 2020-2024: (i) στατιστικά εποχικά μοντέλα SARIMA/SARIMAX, (ii) μη-γραμμικά ensemble δέντρων Random Forest, (iii) gradient-boosted δέντρα XGBoost και (iv) βαθιά επαναληπτικά νευρωνικά δίκτυα LSTM. Η μελέτη επεκτείνεται σε ανασκόπηση βιβλιογραφίας, συστηματική κατασκευή και καθαρισμό δεδομένων, επιλογή υπερπαραμέτρων και διεξοδική ανάλυση σφαλμάτων.

Βιβλιογραφική Ανασκόπηση

Η διεθνής ερευνητική δραστηριότητα γύρω από την πρόβλεψη τιμών ηλεκτρικής ενέργειας είναι ιδιαίτερα έντονη τις τελευταίες δεκαετίες, με εκατοντάδες δημοσιεύσεις. Η βιβλιογραφία καταδεικνύει ότι τα υφιστάμενα μοντέλα μπορούν να ομαδοποιηθούν σε τέσσερις μεγάλες κατηγορίες, καθεμία εκ των οποίων υιοθετεί διαφορετική φιλοσοφία.

1. Παραδοσιακά Στατιστικά Μοντέλα. Βασίζονται σε γραμμικές υποθέσεις και σε θεωρία στάσιμων χρονοσειρών. Χαρακτηριστικά παραδείγματα αποτελούν τα ARIMA/SARIMA και GARCH μοντέλα. Τα μοντέλα αυτά εκτιμούν παραμέτρους μέσω μεθόδων μέγιστης πιθανοφάνειας ή ελαχίστων τετραγώνων και ενώ προσφέρουν διαφάνεια και απλές επεξηγήσεις, έχουν περιορισμένη ικανότητα αποτύπωσης μη-γραμμικών και ξαφνικών αιχμών.
2. Μοντέλα Μηχανικής Μάθησης. Θεωρούν το πρόβλημα ως άσκηση παλινδρόμησης χωρίς αυστηρές υποθέσεις κατανομής. Περιλαμβάνουν δέντρα αποφάσεων, Random Forest, Gradient Boosting/LightGBM, και Support Vector Machines ανάμεσα σε άλλα. Τα αλγοριθμικά δέντρα, διατίθεται για χειρισμό ισχυρά μη-γραμμικών σχέσεων με περιορισμένη ανάγκη προ επεξεργασίας.
3. Βαθιά Μάθηση - Νευρωνικά Δίκτυα. Από τα κλασσικά πολυστρωματικά Perceptron έως τα πιο πρόσφατα RNN, LSTM/GRU που προσπαθούν να μάθουν πολύπλοκα πρότυπα απευθείας από τα δεδομένα. Προσφέρουν θεωρητικά απεριόριστη εκφραστική δύναμη αλλά απαιτούν μεγάλους όγκους δεδομένων, προσεκτική τακτοποίηση υπερ-παραμέτρων και μεγάλους υπολογιστικούς πόρους.
4. Υβριδικές Μέθοδοι. Συνδυάζουν στοιχεία από τις παραπάνω κατηγορίες με στόχο να εκμεταλλευτούν τα συμπληρωματικά πλεονεκτήματα και να μετριάσουν τα μειονεκτήματα κάθε μεμονωμένης προσέγγισης.

Κατόπιν εκτενής μελέτης των εφαρμογών διάφορων μεθόδων σε διάφορες αγορές και χώρες, στο πλαίσιο της παρούσας εργασίας επιλέχθηκε ένα αντιπροσωπευτικό υποσύνολο μοντέλων ανά κατηγορία. SARIMA/SARIMAX εκ μέρους των παραδοσιακών στατιστικών μεθόδων, Random Forest και XGBoost εκ μέρους των αλγορίθμων μηχανικής μάθησης και LSTM ως εκπρόσωπος των νευρωνικών δικτύων.

Δεδομένα και Προεπεξεργασία

Για την εκπαίδευση και αξιολόγηση των υποψήφιων μοντέλων συγκροτήθηκε έναν ενοποιημένο σύνολο δεδομένων βασισμένο στο ENTSO-E Transparency Platform API που καλύπτει την περίοδο 01-01-2018 έως 31-12-2024 και περιέχει τα παρακάτω δεδομένα σε ωριαία συχνότητα.

- Day-ahead Market ωριαία τιμή
- Day-ahead ωριαία πρόβλεψη φορτίου
- Day-ahead ωριαία πρόβλεψη παραγωγής Ηλιακών
- Day-ahead ωριαία πρόβλεψη παραγωγής Αιολικών
- Δείκτης ώρας: παίρνει τιμές 0.0 – 23.0 για τις 24 ώρες της ημέρας
- Δείκτης ημέρας: παίρνει τιμές 0.0 – 6.0 για κάθε ημέρα της εβδομάδας
- Δείκτης εργάσιμης: παίρνει τιμή 1.0 τις εργάσιμες ημέρες, 0.0 αλλιώς

Αφού έγινε η απαραίτητη επεξεργασία των δεδομένων συμπληρώνοντας ελλείψεις εγγραφές, καθορίζοντας 24 εγγραφές ανά ημέρα αντιμετωπίζοντας την αλλαγή της ώρας και προσαρμόζοντας τις εγγραφές σε τοπική ώρα, έγινε ο διαμοιρασμός των δεδομένων σε σετ εκπαίδευσης, 01/01/2020 – 31/12/2023, και σετ δοκιμών, όλο το 2024, κρατώντας μία αναλογία 80/20.

Μεθοδολογία

Για την αξιολόγηση των μεθόδων υλοποιήθηκε rolling-origin στρατηγική όπου κάθε μοντέλο εκπαιδεύεται με δεδομένα μέχρι το τέλος του 2023, στην συνέχεια προβλέπει 24 ώρες μπροστά (επόμενη ημέρα) και έπειτα χρησιμοποιεί τις πραγματικές τιμές που καταγράφηκαν για τις 24 αυτές ώρες για να εκπαιδεύσει είτε από την αρχή (Random Forest, XGBoost), είτε να προσαρμόσει το είδη υπάρχον μοντέλο στα νέα δεδομένα (SARIMAX, LSTM).

Τα διάφορα μοντέλα υλοποιήθηκαν ως εξής:

Για το SARIMA επιλέχθηκαν οι παράμετροι $(1,1,1) \times (1,0,1)_{24}$, ένας συνδυασμός που προέκυψε από τα ACF και PACF plots της πρώτης τάξης διαφοροποιημένης χρονοσειράς που συγχρόνως ελαχιστοποιούν το Akaike Information Criterion (AIC).

Για την Random Forest επιλέχθηκε η δημιουργία 100 τυχαίων δέντρων με μέγιστο βάθος του καθενός ίσο με 10. Για την μέθοδο XGBoost ο αριθμός των δέντρων επιλέχθηκε να είναι 500 ενώ και εδώ το μέγιστο βάθος είναι 10. Επιπλέον εδώ

επιλέχθηκε ο ρυθμός εκπαίδευσης να είναι 0.05 και ο στόχος βελτιστοποίησης το Root Mean Squared Error (RMSE). Και οι δύο μέθοδοι δέντρων χρησιμοποιούν τα ίδια lags για την δημιουργία του εποπτευόμενου πίνακα μάθησης (Lag-1, -24, -48, -168).

Για την υλοποίηση της LSTM χρησιμοποιήθηκε ένα μήκος ακολουθίας 168 και μετά από πολλές δοκιμές και συνδυασμό παραμέτρων επιλέχθηκε η αρχιτεκτονική με 1 στρώμα 24 κυττάρων, Dropout στρώμα 20%, Batch Normalization στρώμα και 24 νευρώνες εξόδου. Σχετικά με την εκπαίδευση επιλέχθηκε μέγεθος batch 32, ρυθμός εκπαίδευσης 0.001 και 200 epochs για εκπαίδευση με διαθέσιμη πρόωρη διακοπή (patience = 10).

Για την αξιολόγηση και σύγκριση των μεθόδων χρησιμοποιήθηκαν τα σφάλματα: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Symmetric Mean Absolute Percentage Error (sMAPE), και Normalized MAE (nMAE).

Αποτελέσματα

Η αξιολόγηση των μεθόδων στο σύνολο του 2024 κατέδειξε ξεκάθαρη ιεράρχηση και νικητήριο μοντέλο για την πρόβλεψη της τιμής της ηλεκτρικής ενέργειας της Ελλάδας στην DAM. Οι μέθοδοι αυτοί συγκρίθηκαν με μια πρότυπη Persistence μέθοδο η οποία δεν κάνει υπολογισμό παραμέτρων αλλά πραγματοποιεί την πρόβλεψη χρησιμοποιώντας ένα παράθυρο επτά ημερών ανάλογα την ημέρα της εβδομάδας και υπολογίζει μέση τιμή των παρελθοντικών τιμών για κάθε ώρα της επόμενης ημέρας.

Η XGBoost υπερείχε συστηματικά, επιτυγχάνοντας μείωση του MAE κατά 43% και RMSE 33% έναντι της Persistence μεθόδου καθιστώντας την απόσταση στατιστικά σημαντική. Η Random Forest ακολούθησε σε κοντινή απόσταση παρουσιάζοντας υψηλότερο σφάλμα αλλά ανθεκτικότητα σε ημέρες με πολύ χαμηλές τιμές. Ένα βέλτιστα ρυθμισμένο SARIMAX, εμπλουτισμένο με όρους Fourier αντί για τις ημερολογιακές μεταβλητές του dataset, μαζί με τις εξωγενείς μεταβλητές πρόβλεψης, περιόρισε το σφάλμα του κατά ~19% έναντι του benchmark αλλά παρέμεινε λιγότερο ακριβές από την XGBoost. Η LSTM με ακολουθία εισόδου 168 ωρών και 64 νευρώνες δεν ξεπέρασε την επίμονη πρόβλεψη παρά μόνο μετά την εισαγωγή των ημερολογιακών δεικτών και εξωγενών προβλέψεων φορτίου και ανανεώσιμων πηγών ενέργειας. Ακόμη κι έτσι, η βελτίωση ήταν οριακή και όχι σταθερή για όλους τους μήνες του έτους.

Η προσθήκη φυσικών ερμηνεύσιμων εξωγενών μεταβλητών (day-ahead φορτίου, αιολικά και φωτοβολταϊκά forecasts) αποδείχθηκε καθοριστική, καθώς σε όλες τις κατηγορίες το MAE μειώθηκε κατά 15-22% σε σχέση με εκδόσεις χωρίς εξωγενείς μεταβλητές. Ιδιαίτερο ενδιαφέρον έδειξαν οι ώρες 19:00 – 22:00 όπου οι δένδρο-βασιζόμενες μέθοδοι προέβλεπαν μεν τον χρόνο εκδήλωσης της αιχμής αλλά και πάλι δεν μπορούσαν να πετύχουν ακριβώς το πλάτος της πραγματικής τιμής ενισχύοντας

την παρατήρηση πως οι όλες οι μέθοδοι δυσκολεύονται, άλλες περισσότερο και άλλες λιγότερο, τις ώρες που παρουσιάζονται οι μη γραμμικότητες και οι απότομες αιχμές.

Μελλοντικές Επεκτάσεις

Η παρούσα εργασία δείχνει ξεκάθαρα αποτελέσματα για την βραχυπρόθεσμη πρόβλεψη στην ελληνική Αγορά Επόμενης Ημέρας. Παρά τα ισχυρά αποτελέσματα, ο ταχέως μεταβαλλόμενος ενεργειακός τομέας δημιουργεί νέες απαιτήσεις και προσφέρει γόνιμο έδαφος για περαιτέρω έρευνα. Διάφορες βασικές κατευθύνσεις είναι τα πλουσιότερα θεμελιώδη δεδομένα με την ενσωμάτωση τιμών TTF gas, διασυνοριακή δυναμικότητα, και τιμές εξισορρόπησης. Επιπλέον, φυσική συνέχεια αποτελεί η μελέτη προηγμένων υβριδικών και πιθανοτήτων μοντέλων σε συνθήκες της Ελληνικής αγοράς καθώς και η προσαρμογή στο νέο SDAC και την 15-λεπτή ανάλυση.

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1 Introduction

1.1 Energy Transition and Challenges

The term “energy transition” refers to the ongoing structural transformation of the global energy sector from a system heavily reliant on fossil fuels to one built on sustainable, decarbonized, and increasingly decentralized energy sources. This shift is driven by mounting concerns over climate change and ambitious international and regional emissions-reduction targets, alongside rapid advances in renewable energy and storage technologies and evolving patterns of energy consumption. Historically, industrialized economies have depended on coal, oil, and natural gas to power growth, but climate imperatives and sustainability concerns are now driving a fundamental shift away from that paradigm. Governments and industries worldwide are mobilizing to achieve this transformation, acknowledging that decarbonizing power systems is essential for meeting international climate commitments and ensuring long-term energy security and resilience. In Europe, the transition holds particular significance: the EU has set a legally binding goal of climate neutrality by 2050, and member states are rapidly expanding renewable generation capacity and modernizing their electricity grids to integrate decentralized generation. This ongoing shift represents one of the most significant reconfigurations of the energy landscape in modern history, with wide-ranging implications for economies, societies, and the environment.

1.1.1 The 3 D’s of the Energy Transition

To make this transition feasible and achievable, there are 3 key dynamics that drive the desired outcome, Deregulation, Decarbonization, and Decentralization, often dubbed as the “3 D’s of the energy transition.”

Deregulation refers to the opening of formerly state-controlled power systems to competition and market forces. Deregulation began in the late 1970s from Chile and the USA. Moving on, the UK’s power market reforms in the late 1980s and continuing through successive European Union directives, deregulation has unbundled vertically

integrated utilities into separate generation, transmission, and distribution entities, introduced competition in generation and retail supply, and replaced rigid regulation with market-based pricing. This shift has enabled new entrants and innovative business models in electricity supply.

Decarbonization reflects the policy-driven move away from fossil fuels toward low- and zero-carbon energy, in response to climate change commitments. The EU has set binding targets for cutting greenhouse gas emissions (e.g., a net zero economy by 2050), and member states are rapidly expanding Renewable Energy Sources (RES) (wind, solar, etc.) and low-carbon technologies. In addition to RES, decarbonization includes the adoption of cleaner fuels (such as green hydrogen and biogas), the electrification of sectors like transport and heating, and the implementation of carbon pricing and emissions trading schemes. Governments and industries are increasingly investing in technologies that support energy efficiency, carbon capture and storage (CCS), and demand-side management to reduce reliance on fossil fuels. Furthermore, financial institutions are adjusting their portfolios to prioritize green investments, aligning capital with climate goals. Altogether, decarbonization is a comprehensive transformation that affects power generation, consumption behavior, industrial processes, and regulatory frameworks, aimed at achieving a sustainable and resilient energy future.

Decentralization denotes a structural change towards distributed energy resources (DER), which is generation that is located close to load (rooftop solar, small wind farms, combined heat-and-power, microgrids) rather than large, centralized plants. Distributed generation lowers transmission losses and often allows consumers to become “prosumers” of energy. But decentralization also involves the integration of energy storage (such as residential batteries), smart meters, local energy markets, peer-to-peer trading platforms, and advanced grid technologies that allow real-time management and optimization of distributed assets. It represents a shift from a top-down energy model to a more flexible, bidirectional system where energy flows between users and the grid. This structural transformation enhances grid resilience and enables tailored local energy solutions. Moreover, decentralization empowers communities and municipalities to pursue energy autonomy through initiatives like community solar programs, local energy cooperatives, and demand-response schemes. It also complements decarbonization by facilitating the integration of intermittent renewables and reducing the need for long-distance transmission

infrastructure. As a result, decentralization is a critical enabler of both democratized energy access and the broader transition to a low-carbon economy.

These three forces together are reshaping the electricity landscape: for example, deregulated markets provide entry points for decentralized generators, while decarbonization drives the deployment of renewable DER units across the grid. Their interplay forms the backbone of a modern, sustainable, and adaptive energy system capable of meeting the environmental, economic, and social challenges of the 21st century.

1.1.2 The Main Axes of change in Deregulated Markets

The electricity industry, in general terms, has four main functions: generation, transformation, distribution and supply of the electric power. Historically, a single firm, primarily a government entity, was given the mandate to oversee all four functions leading to a legally endowed monopoly. This state-owned national power entity, was responsible for the minimization of the costs, the coordination between this functions and financing the development of the power systems. However, over time, this model, under the monopoly market, revealed structural inefficiencies. It often lacked the necessary incentives for cost reduction, service quality improvements, technological innovation, and consumer responsiveness. Moreover, the absence of competition meant there was little pressure on utilities to operate efficiently or adapt to evolving demand patterns and environmental goals. The deregulation of the electricity markets seeks to address these shortcomings [1] [2].

The Deregulation (or liberalization) of electricity markets worldwide had generally followed three main axes:

Privatization: State-owned utilities have been divested to private ownership. Privatization is intended to bring in capital and managerial expertise, improving efficiency. In many countries, governments sold off generation and distribution assets, and as a result, independent power producers (IPPs) now account for a large share of generation in liberalized markets.

Competition: Electricity markets have been restructured to allow competition in generation and supply, while recognizing that transmission and distribution grids remain natural monopolies due to their high infrastructure costs and physical con-

straints. Under traditional monopoly structures, captive customers had no choice of supplier. Deregulated markets, however, introduced wholesale competition among generators and retail competition among suppliers. These reforms aim to reduce prices, improve service quality, and encourage innovation by making providers compete for customers. In addition, competition has reduced the benefits traditionally attributed to large monopoly suppliers and created an opportunity for smaller firms to compete and bring niche solutions and RES into the mix, helping to diversify energy sources.

Unbundling: Each firm can only participate in one of the four segments of the industry. The natural monopoly segments (transmission and distribution networks) are separated from potentially competitive segments. Unbundling ensures that network operators do not favor affiliated generators or suppliers. This reform involves breaking up an integrated utility into multiple generation and distribution entities that trade with each other in a competitive wholesale market. Full legal or functional unbundling of grids is a standard requirement in EU electricity market directives.

1.2 Overview of European Electricity Market

Electricity markets are typically divided into two main segments: wholesale and retail. Each segment plays a distinct role in the overall functioning of the power system and involves different sets of actors, contracts, and regulatory frameworks.

The wholesale market is where electricity is bought and sold in large quantities, typically by generators, retailers, large industrial consumers, and sometimes traders or aggregators. This market operates on various timescales and is often managed by a centralized operator such as an Independent System Operator (ISO) or a Transmission System Operator (TSO). Wholesale prices fluctuate based on supply and demand conditions, fuel costs, availability of renewables, and transmission constraints. The wholesale market ensures that enough generation is scheduled to meet demand and provides price signals for investment in generation capacity and flexibility.

The retail market, on the other hand, serves as the interface between electricity suppliers (also known as retailers or utilities) and end-use consumers, including households, commercial enterprises, and small- to medium-sized industries. In

this segment, retail suppliers purchase electricity from the wholesale market and resell it to final consumers, often bundling energy supply with customer services such as billing, metering, and energy efficiency programs. Retail transactions are governed by contracts and tariffs, which are determined competitively, allowing consumers to choose among various offers based on price, renewable content, or service features. liberalized or competitive retail markets allow multiple suppliers to compete for customers, fostering innovation, product differentiation, and often lower prices. Customers in such markets may select contracts with fixed or variable pricing, green energy options, or even dynamic tariffs linked to real-time wholesale prices.

1.2.1 Wholesale Market Participants

Focusing on the wholesale electricity market, as said earlier, this is where bulk power is bought, sold and scheduled for delivery before reaching end users. Transactions in the wholesale market often occur in structured intervals and are essential for balancing supply and demand on a continuous basis. A wide range of participants operate within this space, each playing a crucial role in the overall functioning of the electricity sector.

At the heart of the wholesale market are the **power producers**, who own and operate power plants using various energy sources such as fossil fuels (coal, natural gas), nuclear energy, hydroelectric power, wind turbines, and solar photovoltaic systems. These entities produce the electricity that is injected into the transmission grid and made available for trading.

Wholesale suppliers are another key group. These are often energy companies or financial trading firms that purchase electricity from generators and then resell it, either to other market participants or directly to retail suppliers. They actively participate in market bidding processes by submitting buy and sell orders through organized exchanges. Their role is not only to facilitate energy flow but also to manage price risks, optimize portfolios, and sometimes provide liquidity to the market.

Transmission System Operators (TSOs) are responsible for maintaining the stability and reliability of the high-voltage grid. Their tasks include real-time balancing of

supply and demand, managing congestion, and enabling the secure transmission of electricity over long distances. TSOs also facilitate market coupling, which enables cross-border electricity trading within regional markets (such as the European Internal Electricity Market) and ensures that electricity flows from areas of lower price to areas of higher demand in a cost-efficient manner.

The **market operators**, typically known as power exchanges, play a critical institutional role. These entities, such as EPEX SPOT, Nord Pool, or HEnEx (the Hellenic Energy Exchange in Greece), organize the actual trading platforms where buyers and sellers submit their bids and offers. These exchanges aggregate all orders, run auction mechanisms, match supply and demand, and determine the market clearing price and clearing volume. The transparent and standardized operation of these exchanges helps establish trustworthy and reference wholesale electricity prices, which are widely used by market participants and regulators alike.

Additionally, **Distribution System Operators (DSOs)**, although primarily focused on managing electricity flows at the local and medium/low-voltage level, interact with the wholesale market indirectly. DSOs help implement distributed energy resource integration and support demand response schemes. Alongside them, regulators as well as ancillary service providers, contribute to the broader framework that ensures system security, fair market practices, and long-term reliability.

1.2.2 Wholesale Market Operation

Price formation on the EU Day-Ahead spot market is fundamentally rooted in the principle of marginal cost pricing. Marginal cost pricing, in economics, is the practice of setting the price of a product to equal the extra cost of producing an extra unit of output [3]. According to [4], electricity prices are determined based on the cost of producing one additional unit of electricity, specifically, one megawatt-hour (MWh).

In European wholesale power markets, this principle becomes the foundation of day-ahead electricity trading in Europe. Most wholesale electricity trade in the EU is conducted through centralized market platforms, commonly known as power exchanges, where participants engage in Day-Ahead Market (DAM) auctions. These markets operate on a 24-hour rolling schedule, meaning that electricity for each hour

of the following day is auctioned on the day before delivery. Participants, including generators, retailers, and sometimes large industrial consumers, submit supply offers (how much electricity they are willing to produce and at what price) and demand bids (how much they are willing to consume and pay for it) for the upcoming day. At present, bids and offers are submitted on an hourly basis, meaning the market determines one price for each of the 24 hours of the following day. However, this structure is set to evolve. In response to the growing share of variable renewable energy and the need for increased system flexibility, the EU and power exchanges are preparing to shift toward 15-minute market resolution. This change, already implemented in intraday markets, is expected to be extended to the day-ahead market in the near future.

Once all bids and offers are collected, the exchange platform organizes them into aggregate supply and demand curves. The supply curve is built by ordering all generation offers from the lowest to the highest price, a process known as merit-order stacking. Conversely, the demand curve consists of all consumption bids, sorted from highest to lowest price. The market clearing mechanism determines the market clearing price and volume by finding the intersection point of these two curves. This is the point where the total quantity of electricity supplied matches the total quantity demanded, and it sets the uniform price for that hour.

This system is known as a pay-as-clear or uniform pricing model. It means that all sellers whose offers are accepted, regardless of their individual bid price, are paid the same market clearing price. Infra-marginal generators, whose costs are lower than the clearing price, receive a surplus or “profit” above their production cost, while infra-marginal bids have positive “utility” as the price is less than the marginal benefit. These profits are often necessary to recover fixed and capital costs not covered by marginal cost alone. On the other hand, extra-marginal offers, whose prices exceed the clearing price, are rejected, as they are not needed to meet demand for that hour. In essence, the last accepted unit of electricity, also called the marginal unit, is the most expensive offer needed to satisfy total demand, and it determines the clearing price for everyone [5].

Graphically, the cleared volume is the quantity at which the ascending supply curve meets the descending demand curve. The resulting price-volume point maximizes total economic welfare, i.e. the sum of consumer and producer surplus, under the given supply and demand curves. In fact, modern clearing algorithms explicitly solve

for the allocation that maximizes this social welfare (economic surplus) subject to system constraints [6]. The marginal cost pricing system that was described, along with the market equilibrium point can be seen in the simplified Figure 1.

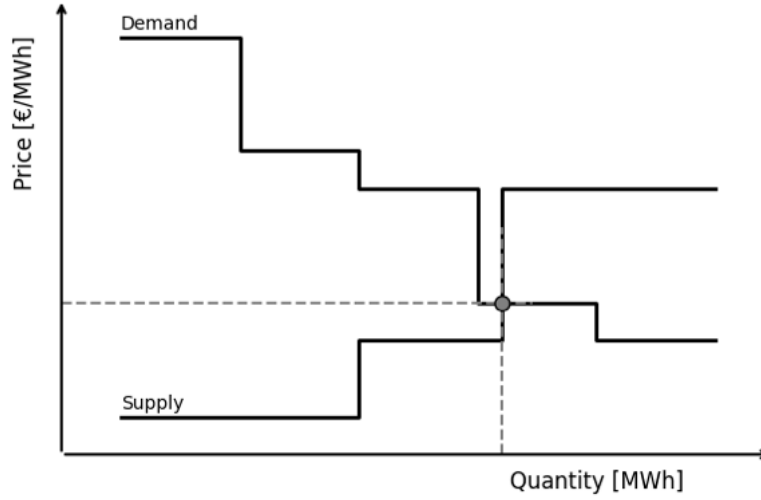


Figure 1: Marginal Cost Pricing.

In practice the actual operation of the grid involves a much more complex optimization process. A host of technical, operational, and network-related constraints must be taken into account to ensure that this economic dispatch is physically feasible and secure for the power system.

On the generation side, there are several technical limitations of power plants that might not make them dispatchable at all times even though they might have the lowest marginal cost. Beginning with ramping constraints, they refer to the maximum rate at which a generating unit can increase or decrease its output. This is especially relevant for thermal plants (like coal or gas units), which cannot instantly change their output due to thermal and mechanical limitations. In addition, minimum up and down times denote the minimum amount of time a power plant must stay online once it has been turned on, or remain offline after it has been turned off. Finally, minimum stable generation levels indicate the lowest output a generator can maintain while operating stably.

On the transmission side, the electricity grid is subject to physical flow limitations that further complicate dispatch decisions. Electricity does not flow like a commodity through pipelines or trucks as it follows Kirchhoff's laws, which means it distributes itself along all available paths in inverse proportion to resistance. However, trans-

mission lines have finite thermal and stability limits, and exceeding these limits can lead to overheating, voltage instability, or even cascading failures and blackouts. To comply to this constraints, Europe follows a zonal pricing model, where countries (or large regions within them) are divided into bidding zones. Each zone is treated as a single price area in the day-ahead market. Within each zone, it is assumed that there are no internal constraints, and all accepted buy and sell orders share the same uniform clearing price. However, cross-border power flows between zones are restricted based on Available Transfer Capacities (ATCs), the maximum amount of power that can be safely transferred across inter-connectors without jeopardizing grid security. These ATCs are calculated in advance by the TSOs, who assess the state of the grid, forecasted flows, and contingencies. They serve as essential input to the market clearing process, as they define how much power can realistically be exchanged between different regions.

Because of the interplay between unit-level constraints and cross-zonal transmission limits, clearing the electricity market becomes a complex mathematical problem, specifically, a constrained optimization that must balance economic efficiency with physical feasibility. This problem is solved by a market integration algorithm called EUPHEMIA [6].

1.2.3 Electricity Markets and European Target Model

Electricity, unlike most other commodities, cannot be economically stored in large quantities, making it essential that supply and demand are continuously balanced in real time [7]. However, the inherent uncertainty associated with electricity demand, generation outages, and especially the variability of renewable energy sources (RES) such as solar and wind, introduces significant complexity to this task. To address these challenges, electricity markets in Europe are structured into multiple timeframes, each with distinct characteristics, time horizons, and operational objectives. These markets provide a platform for participants to gradually resolve uncertainty as real-time operation approaches, enhancing both economic efficiency and system security. The principal market timeframes in the European electricity system include:

Forward Markets: These markets allow market participants (e.g., suppliers, large consumers, traders) to hedge against future price volatility by entering into contracts weeks, months, or even years in advance. Forward products are typically over-the-

counter (OTC) or traded on exchanges such as the European Energy Exchange (EEX). Although not physically binding, forward contracts serve as important financial tools to manage long-term price risk.

Day-Ahead Market (DAM): The DAM is the cornerstone of the European wholesale electricity market. Participants submit their offers to sell and bids to buy electricity for each hourly period of the next day. This market closes at 12:00 CET the day before delivery. The matching process aims to maximize social welfare by solving a market-clearing optimization problem that determines the accepted volumes and a uniform market-clearing price per bidding zone. The Day-Ahead market provides the first physically binding schedule and forms the main reference price for electricity.

Intraday Market (IDM): As new information becomes available (e.g., updated RES forecasts, unit availability), the Intraday Market allows participants to adjust their positions closer to real time. Trading in the intraday market starts after the publication of DAM results and continues until shortly before real-time (as close as 5 minutes in some regions). In Europe, continuous trading and intraday auctions are both used. The IDM plays a critical role in improving forecast accuracy and enhancing the efficiency of the real-time dispatch.

Balancing Market: Despite all prior market layers, deviations between supply and demand still occur due to forecast errors or unexpected events. Transmission System Operators (TSOs) activate balancing reserves to restore real-time balance. Market participants must submit balancing capacity and energy offers, which are used based on merit order, system needs, and reserve availability. Settlement in the balancing market is typically based on imbalance prices, which provide incentives for accurate forecasting and scheduling. Together, these sequential markets help reduce uncertainty in stages, allowing participants to continuously revise their expectations and hedging strategies while maintaining a secure and cost-effective operation of the power system.

Ancillary services markets: To support balancing, TSOs maintain ancillary services markets where reserve capacity is procured. In the European framework these reserve markets are separate from the energy spot markets. TSOs announce reserve requirements (capacities for frequency control, etc.) and purchase them via auctions or tenders, aiming to maintain a precise frequency (50 Hz in Europe). The key reserve products include [8]:

- **Frequency Containment Reserve (FCR)** – Primary control. This is the fastest reserve: fully automatic and symmetric (can increase or decrease output). FCR kicks in within seconds of a frequency deviation to arrest the disturbance. It provides a short burst of power (typically up to 15 minutes) to stabilize frequency around 50 Hz.
- **Automatic Frequency Restoration Reserve (aFRR)** – Secondary control. This reserve is also automatic but slower than FCR: it is activated (triggered) within about 5 to 10 minutes and adjusts output for up to about 12.5 minutes.
- **Manual Frequency Restoration Reserve (mFRR)** – Tertiary control. This is a slow reserve activated manually by the grid operators to address longer-lasting imbalances, with a typical activation of 30 to 60 minutes.

The **European Target Model (ETM)** is a policy framework and set of rules aimed at creating an integrated pan European electricity market across all timeframes. Its origins lie in the EU's Third Energy Package (2009), which empowered the Agency for the Cooperation of Energy Regulators (ACER) and mandated unbundling and cross-border cooperation. Building on this, ACER and ENTSO-E developed network codes that define the target model's building blocks. In essence, the ETM requires implicit capacity allocation (market coupling) in day-ahead and intraday markets and harmonized balancing rules, so that electricity and cross-zonal capacity are allocated together [9]. Under the target model: market areas (bidding zones) use joint auctions and products and gate times are standardized. The objectives are to maximize cross-border welfare, remove market fragmentation, support competition and renewables integration. For example, implicit allocation (i.e. joint energy and capacity auctions) is the mandatory European approach for day-ahead/intraday markets [10]. In practice, this means a single order book and clearing algorithm for all coupled zones, ensuring that capacity goes to the trades with the highest value (equalizing price differentials). The overall aim is a single internal market where power flows freely, price signals reflect system conditions, and market participants face the same rules regardless of borders.

Many elements of the ETM are now in place. The **Single Day-Ahead Coupling (SDAC)** has been fully implemented since 2014: 27 countries clear their day-ahead markets jointly via the EUPHEMIA algorithm. More recently, **Single Intraday Coupling (SIDC)**, built on the XBID platform, was launched in 2018 and now

spans most EU countries. This allows continuous cross-border intraday trading up to the gate (typically 1 hour before delivery) [11]. These couplings have significantly increased liquidity and allowed more efficient use of interconnectors. Balancing and reserve market integration is also underway under the European Balancing Regulation (2017). All TSOs must join common platforms: IGCC was launched in 2016 to implement the imbalance netting process, TERRE (for replacement reserve) in 2019, and as of 2022 both the PICASSO (automated aFRR) and MARI (manual mFRR) platforms are live. Together, PICASSO/MARI/TERRE/IGCC complete the target design for balancing markets [12]. These platforms standardize products (e.g. 15-minute aFRR blocks, pay-as-cleared pricing) and enable cross-border sharing of reserve capacity, increasing overall system flexibility. In summary, through SDAC, SIDC/XBID, and the emerging guideline on electricity balancing (EBGL) platforms, the EU is progressively realizing the target model across all timeframes (day-ahead, intraday and balancing) as envisaged by ACER/ENTSO-E and EU legislation.

1.3 Greek Electricity Market

Greece, like many other countries, originally operated under a vertically integrated, monopolistic electricity system dominated by a single public entity, Public Power Corporation (PPC). Over the past two decades, however, the country has undergone a progressive process of deregulation and market liberalization, aiming to align with European Union directives and create a more competitive, transparent, and efficient electricity market. This transition laid the groundwork for the establishment of organized wholesale markets, the unbundling of system operations, and the integration of private actors into the energy landscape.

Greece has now fully aligned its wholesale electricity market with the EU Target Model. A new market design was launched on 1 November 2020 by the Hellenic Energy Exchange (HEnEx). On that date the day-ahead market (DAM) was opened (with market coupling initially in “isolated” mode, i.e. explicit interconnector allocation), followed on 2 November 2020 by the intraday market (IDM) and the balancing market. Under this reform, all generators (including renewables and CHP) must bid into HEnEx’s markets (with smaller RES plants aggregated via DAPEEP) rather than a mandatory pool. Subsequently, Greece connected these markets to its

neighbours: the day-ahead market was coupled with Italy via SDAC on December 2020 and with Bulgaria in March 2021 [13]. A continuous intraday market (24/7 trading through the SIDC platform) began on November 2022. Looking ahead, IPTO (Independent Power Transmission Operator) will integrate Greek reserve markets into European balancing platforms: joining the PICASSO aFRR platform on 2025 and the MARI mFRR platform by 2026.

The main market operators and roles in Greece are the following:

- **HEnEx (Hellenic Energy Exchange S.A.)** - Greece's Nominated Electricity Market Operator (NEMO). HenEx operates the wholesale Day Ahead and Intraday trading platforms in the Greek bidding zone. It provides the matching engine (hourly DAM and intra-day auctions/trading) and supports the following products for trading: Hourly Hybrid Orders, Block Orders, Linked Block Orders, Exclusive Group of Block Orders for the DAM market and Hourly Hybrid Orders for the Intraday market.
- **IPTO/ADMIE (Independent Power Transmission Operator)** - the national Transmission System Operator. IPTO is responsible for operating, controlling, maintaining and developing Greece's high voltage grid. It runs the Balancing Market and real-time dispatch (after the integrated day ahead scheduling) and procures reserve capacity (FCR, aFRR, mFRR). IPTO sets grid schedules and reserve requirements (via an Integrated Scheduling Process), and it allocates cross border transmission capacity.
- **DAPEEP (Renewable Energy Sources Operator & Guarantees of Origin)** - the operator for RES and CHP market integration. DAPEEP manages the support schemes and settlement accounts for renewable generation and is the official registry for Guarantees of Origin (GOs) for renewable electricity. It organizes public auctions and support contracts under the national law. Until all renewables assume balancing responsibility, DAPEEP acts as aggregator of last resort for small RES/CHP producers.
- **RAE/RAAEY (Regulatory Authority for Energy, now Energy, Waste & Water)** - the independent energy regulator. RAE oversees the electricity markets, issues licenses, sets network tariffs and market rules, and enforces EU regulations. It reviews market operation (issuing wholesale market re-

ports), approves grid codes and market designs (such as the Target Model implementation), and ensures non-discriminatory access.

- **HEDNO (Hellenic Electricity Distribution Network Operator S.A.)** - the sole distribution system operator. Established in 2011 to unbundle distribution from PPC, HEDNO owns and maintains the medium- and low-voltage network across Greece. Its duties include developing and operating the distribution grids (including islands' grids) under RAE supervision. Importantly, HEDNO also runs the markets of the non interconnected islands (NII), i.e. it administers generation dispatch and settlement for island grids not physically linked to the mainland. HEDNO handles all small generator connections and ensures reliable service to end customers on distribution networks.

1.4 Purpose of this Thesis

This thesis, focuses on electricity price forecasting, which refers to the quantitative estimation of future market-clearing prices before they are actually released. In the DAM these hourly prices act as the principal signal of marginal generation cost and, by extension, of system scarcity. Accurate forecasts are therefore indispensable to virtually every stakeholder in the power sector. Generators and retailers rely on them to optimize bidding strategies, schedule units, and hedge portfolio risks, while traders use them to assess reserve sufficiency and spread products. TSOs require them to assess reserve sufficiently and congestion management, regulators employ them in market monitoring and long-term adequacy studies and investors need credible trajectories to evaluate asset profitability. In short, reliable DAM price forecasts translate directly into lower operating costs, improved risk management, and more efficient allocation of capital.

Providing such forecasts has become more difficult and simultaneously more valuable over the last years. The ongoing energy transition is transforming the generation mix toward weather-dependent renewables, electric-heat decarbonization and distributed resources, all of which increase net-load volatility. At the same time, the recent global energy crises have introduced unprecedented external shocks. The COVID-19 pandemic led to sharp, asynchronous swings in electricity demand, while Russian invasion of Ukraine in 2022 precipitated dramatic increases and volatility in European

natural-gas and emissions-allowance prices, which cascade into wholesale power prices. These developments have resulted in price series characterized by frequent spikes, regime shifts and pronounced non-stationarity, challenging the assumptions of traditional linear time-series methods and underscoring the need for more advanced modeling approach.

Against this backdrop, this thesis aims to develop and compare several forecasting models for Greek DAM prices, focusing on machine-learning and statistical techniques. Specifically, the study implements and evaluates: ARIMA (Autoregressive Integrated Moving Average), a classic statistical time-series model, Random Forest and XGBoost, tree-based ensemble machine learning regressors, and LSTM (Long Short-Term Memory neural networks), a deep learning model suited for sequential data. Each model is trained on historical price and relevant exogenous variables (such as demand forecasts and renewable generation). The objectives are to implement these famous methods, assess forecasting accuracy under Greek conditions, understand each method's strengths, do a comparative analysis, and identify the most effective approach for short-term (day-ahead) prediction.

1.5 Structure of this Thesis

The thesis is organized into eight chapters that progressively develop the study from background context through to conclusions and findings.

Chapter 1 – Introduction: Presents the motivation and backdrop for electricity price forecasting. It reviews the forces driving the energy transition, outlines the architecture of European wholesale markets, describes the European Target Model and the Greek market, and states the purpose and scope of the study.

Chapter 2 – Literature Review: Summaries previous work on statistical, machine learning, deep learning and hybrid techniques for day-ahead price prediction, highlighting knowledge and addresses the gaps that this thesis addresses.

Chapter 3 - Examined Forecasting Methods: Details the theoretical underpinnings and practical mechanisms of the four models that the thesis investigates (ARIMA, Random Forest, XGBoost, LSTM).

Chapter 4 – Data Collection and Analysis: Describes the assembly of historical prices and exogenous variables, the cleaning and preprocessing steps, feature engineering, explanatory analysis and the construction of train/test splits.

Chapter 5 – Experimental Setup and Implementation: Explains the modeling pipeline, hyper-parameter settings, software environment and evaluation metrics used to ensure a fair and reproducible comparison across methods.

Chapter 6 – Results and Analysis: Presents the forecasting performance of each model, provides the statistical significance tests, and discusses how predictor choices influence accuracy under Greek market conditions.

Chapter 7 – Conclusions and Future Work: Distills the main findings, reflects on their practical implications for market participants and policymakers, and proposes avenues for further research.

Chapter 8 – References: Lists all scientific and regulatory sources consulted throughout the study.

2 Literature Review

Time series forecasting involves predicting the future of a sequence based on its previous observed values and patterns. In electricity price forecasting (EPF), the goal is to model the complex temporal dynamics of market prices, which often exhibit daily and weekly seasonality, yearly cycles, and abrupt spikes, to accurately predict future prices. Unlike stationary financial series, electricity prices are highly volatile and present extreme spikes but are not purely random as they follow recurrent demand supply patterns (e.g. peak hours vs off-peak) and are influenced by many exogenous factors such as weather, fuel costs, regulations etc. Effective forecasting models must capture these patterns and handle challenges like non-stationarity and heteroscedasticity (i.e. changing variance over time).

EPF methods can be broadly grouped into four categories: statistical models, machine learning models, deep learning models and hybrid models [14]. Originally, work on EPF relied heavily on statistical time-series techniques which assume linear relationships and often require the time series to be stationary. Usually, this is not the case for electricity price datasets. Thus, machine learning and deep learning models or often hybrid combinations, have collected extreme popularity in recent years, since they can model non-linear relationships and adapt to complex data distributions.

Wide range of models are proposed in academic papers and each of them has specific characteristics. As the authors of [15] clearly indicate, the EPF literature is heavily fragmented with an overwhelming number of models proposed, tailored to a specific dataset or market context. Some key factors that distinguish each model and EPF attempts are forecast horizons, different strategies, datasets and predictors. Forecast horizons are typically categorized as short-term (hours to days ahead), medium-term (weeks to months), or long-term (year ahead), depending on the application. In addition, two fundamental strategies for multi-step forecasting are often used: direct forecasting, where separate models predict each future time step (or a sequence of steps) directly, and recursive forecasting, where a single-step model's output is fed back as input to predict further ahead. In general, direct multi-step methods can avoid error accumulation, whereas recursive approaches leverage a one-step model for simplicity but risk compounding errors over multiple steps. The other

factors are different datasets, namely the country for which the EPF occurs and the period it concerns (months/years), while the different predictors relate to the different exogenous variables and lagged features the model uses to learn the patterns and make the predictions.

2.1 Statistical Methods

Statistical forecasting models use historical price series, and sometimes exogenous variables, to extrapolate future prices based on identified patterns like trends and seasonality. A foundational statistical approach is the Autoregressive Integrated Moving Average (ARIMA) and its seasonal extension SARIMA, which model price as a linear function of its own past values and error terms.

The authors of [16], investigated and compared methods for forecasting day-ahead electricity prices in Germany. Specifically, they used four different regression models: ARIMA, SARIMA, SARIMAX, Multiple Linear Regression (MLR). These models were evaluated on their ability to predict daily electricity prices using various combinations of endogenous and exogenous variables. ARIMA and SARIMA models relied solely on past electricity prices, while SARIMAX was enhanced to incorporate external predictors and MLR used exogenous variables with additional dummies for weekend effects and trend terms. A preprocessing of data including differencing of data and autocorrelation analysis, among others, was done to determine the parameters of the models. The study concluded that incorporating exogenous variables and seasonality significantly improved accuracy, finding the SARIMA model outperforming the rest and achieving a 4.8 RMSE.

To improve the performance of the ARIMA models on high volatility periods, authors in [17] propose the application of GARCH model for forecasting day-ahead electricity prices in the Spanish and Californian electricity markets. The model combines a standard ARMA structure for mean estimation with a GARCH process for modelling conditional variance, capturing both serial correlation and volatility clustering often presented in volatility prices. The paper shows that GARCH models are highly effective and offer improved accuracy in crisis periods.

Long-term price forecasting using SARIMA, was experimented by the authors of

[18]. Specifically, the paper is utilizing different factors like natural gas, Gross Domestic Product (GDP), and electricity imports, among others, that appear to have high correlation with electricity price. The goal is to directly forecast the average monthly price for the next three or five years of the dataset. The model was also compared with other state-of-the-art machine learning models and appeared to lack in performance as the results indicate.

Overall, statistical models are prized for their simplicity and interpretability, but they struggle with non-linear, regime-switching behavior of power prices like sudden jumps due to outages or weather events. In some cases, ARIMA, GARCH and other statistical models are used as benchmark methods to compare against more complex models.

2.2 Machine Learning & Deep Learning Methods

Machine learning models offer a flexible, data-driven approach that can capture non-linear relationships between inputs (features) and electricity price outputs. Unlike pure time series models, machine learning methods can readily incorporate a variety of features like lagged prices, system load, weather conditions, fuel prices etc., without requiring an explicit parametric form.

With this in mind, authors of [19] proposed a configuration based on Random Forest (RF) model implementation for the prediction of DAM prices for several European countries with interconnected electric market systems. The model uses two sub-forecasters to forecast the load and the RES production, meaning we have a recursive operation. The load forecasting module is a SARIMA model that uses social indicators along with weather and historical load data and is fine-tuned with the necessary correlation analysis. The RES forecasting module is an ARIMA model that uses historical solar, wind and weather data to make the predictions. The provided inputs by these two models along with a set of external market data are considered for the main price forecaster with RF. The developed RF model is fine tuned in each scenario to follow each country's trends and patterns.

The qualities of machine learning models are also present in [20]. In this study a comprehensive forecasting framework tailored to the Turkish electricity market is

presented. A broad spectrum of forecasting models is developed and evaluated, from statistical methods like ARIMA to machine learning algorithms such as Support Vector Machines (SVM), Extreme Learning Machine (ELM) and Extreme Gradient Boosting (XGBoost). The research also includes deep learning architectures like Multilayer Perceptron (MLP), Long Short-Term Memory (LSTM), Bidirectional LSTM, and Convolutional LSTM. The dataset used, covers hourly prices from January 2021 to September 2023, enriched with macroeconomic indicators, time-based features and calendar effects. The study integrates two sophisticated feature selection techniques to refine the set of input variables, thereby improving model interpretability and predictive performance. The results demonstrate that XGBoost consistently outperforms other models, achieving MAPE values as low as 4.42% on volatile days. In contrast ARIMA and SARIMA perform poorly, while deep learning methods show competitive performance but are slightly less consistent across varying conditions.

In a different study [21], a deep analysis of EPF models is applied to the German electricity market from 2020 to mid-2022. The authors structure their approach by initially exploring statistical methods, like the previous ones, such as ARIMA, SARIMA and AR models with weekday dummies. These are then contrasted against more advanced machine learning techniques like XGBoost and various neural networks including MLPs, LSTM and Convolutional Neural Networks (CNN). Notably, they also incorporate deseasonalization via wavelet decomposition to improve model performance. The authors applied their models to different forecasting granularities: daily average prices and hourly prices across various forecasting windows (from a single day up to full months) and under diverse training/test combinations. The results demonstrate that LSTM models consistently outperform all other methods, especially in short- and medium-term forecasts. On the other hand, in this case XGBoost performs better than naive benchmarks, but lags behind LSTM in capturing temporal dependencies and volatility.

As we saw from the previous two studies, deep learning methods, where also introduced in the EPF fields of research and have presented promising performance. Deep learning is a subset of machine learning that focuses on utilizing multilayered neural networks to perform classification, regression, and representation learning. The methods used can be either supervised, semi-supervised or unsupervised. The most modern deep learning models are based on multi-layered neural networks such

as convolutional neural networks (CNNs) and transformers [22]. In the case of EPF we are dealing with supervised learning as the data are labeled and the task involves regression, which is a classic supervised learning problem. Deep learning approaches have been widely adopted in EPF in recent years. The most used deep learning models for electricity prices are recurrent neural networks (RNNs) and especially LSTM and Gated Recurrent Unit (GRU) networks. These models are usually combined with CNNs and create hybrid approaches to forecasting strategies. LSTM and GRU architectures are designed to handle sequential data by maintaining an internal state (memory) that can learn persistence or seasonality over long time lags.

In their work on EPF, the authors of [23] present a data driven approach utilizing a feedforward Artificial Neural Network (ANN) trained via backpropagation to forecast the day-ahead market clearing price. The model is designed to address the inherent nonlinearity and volatility of electricity prices by capturing temporal dependencies and market dynamics through carefully engineered input features. The dataset used, consists of eight months of historical market data, with six months used for training and the remaining two for testing. Each day is divided into 48 half-hourly time slots, yielding high-resolution input sequences. The input vector to the neural network comprises 13 variables: day of the week, time slot of the day, forecasted demand, demand change and a series of lagged price values. The network's architecture itself includes two hidden layers, with 10 neurons in the first layer and 5 in the second, both using the tansig activation function, followed by a linear output layer that produces the predicted price for a single time slot. A logarithmic transformation was applied to cap extreme values 70 \$/MWh, which are later reverted post-processing. The study concludes that while neural networks offer a viable solution for price forecasting in complex markets, their accuracy during volatile periods needs improvement.

On a different study [24], an application for EPF in the Turkish day-ahead market is developed, utilizing Recurrent Neural Networks and specifically a multi-layer (deep) GRU. The authors employ a rolling-window training strategy with three years of historical hourly price data (2013-2015) to forecast each hour of the subsequent day across all of 2016, resulting in a 24-step forecasting scheme repeated daily. The input feature set includes a comprehensive mix of autoregressive lags, along with exogenous variables such as forecasted demand over supply, weighted average

temperature and balancing market prices. The GRU model consists of three stacked layers followed by a fully connected output layer, using ReLU activation and trained with the Adam optimizer and early stopping based on validation loss. Comparative experiments are conducted against shallow and deep ANNs, LSTMs, CNNs, and traditional statistical models including SARIMA, Markov Regime-Switching AR, and SETAR. Performance is evaluated using Mean Absolute Error (MAE), and the proposed three-layer GRU consistently outperforms all alternatives, achieving the lowest MAE of 5.36 €/MWh.

CNN and DNN models are compared with different machine learning methods in [25], for forecasting day-ahead electricity prices across European markets, addressing the high volatility and complexity of these markets. The authors utilize a range of datasets, including both standard and enriched time series from France, Germany, and Belgium, incorporating not only historical prices but also novel exogenous features such as consumption and production forecasts, prices from neighboring countries, Swiss electricity prices, gas prices, and calendar-related dummy variables. The study compares various machine learning models like Support Vector Regressors (SVR), Random Forest, with DNN and CNN, against traditional auto-regression baselines. It finds that enriched datasets significantly enhance model performance, with SVR and DNN consistently outperforming Random Forest and CNN in terms of MAE, sMAPE, and relative MAE.

Another study [15], presents the review of state-of-the-art algorithms. A comprehensive analysis of methods used for EPF are addressed. The authors, propose a framework by evaluating two representative models: the Lasso Estimated AutoRegressive (LEAR) model, a regularized linear regression approach that incorporates feature selection via LASSO, and a DNN optimized with Bayesian methods. Both models are tested on five open-access databases representing diverse electricity markets (Nord Pool, PJM, EPEX-BE, EPEX-FR, EPEX-DE), each spanning six years and including relevant exogenous variables. Results show that LEAR is a highly competitive statistical model, while DNNs, though more flexible, often lack rigorous competitive evaluation. The paper also provides the datasets, pre-trained models, and a dedicated Python toolbox to promote transparency and reproducibility.

2.3 Hybrid Methods

As stated earlier, recent work on EPF has also been focusing on combining methods and creating hybrid models to achieve better performance and adaptability. Hybrid models seek to leverage the strengths of different methods, for example by combining a statistical model to capture linear dynamics with neural network structure network to capture non-linear residual patterns. Moreover transformers, data decomposition strategies, combinations of multiple preprocessing steps, learning algorithms and even optimization techniques may also be utilized in some cases.

Beginning the review on those efforts, the authors of [26], introduce a comprehensive deep-learning framework for EPF, comprising a feed-forward deep neural network, two hybrid recurrent variants that couple either an LSTM or a GRU layer with a dense block, and a dual-branch convolutional network. This models are benchmarked against the field's 27 state-of-the-art statistical and machine learning methods expanded to 98 model configurations in total. Using hourly Belgian EPEX spot prices from 2010 to 2016 together with load and generation forecasts from both Belgium and neighboring France, the study standardizes data pre-processing and applies uniform, automated hyperparameter tuning via the tree-structures Parzen estimator followed by early-stopped Adam optimization. The comparison shows that the DNN attains lowest error (12.34% sMAPE), with GRU- and LSTM- hybrids close behind ($\approx 13\%$) all three significantly outperforming tradition statistical methods. The CNN although less accurate than its Deep Learning peers, still ranks above non deep learning competitors.

Hybrid structured deep neural networks were also explored in [27]. This paper presents a forecasting method combining CNN and LSTM, aiming to predict electricity prices more accurately by leveraging the strengths of CNN for effective feature extraction and LSTM for capturing temporal dynamics. The data that were utilized are from the Pennsylvania-New Jersey-Maryland (PJM) electricity market, clearing prices recorded in 2017, that present volatility and occasional spikes. The model was compared with other different approaches like SVM, Random Forest, MLP, CNN and LSTM individually. The hybrid approach demonstrated superior performance by achieving the lowest error rates.

On a different study [28], the authors used historical electricity market data from the

German electricity market, complemented with climatic conditions and developed multiple deep learning models including a CNN enhanced with an attention mechanism, a hybrid model combining CNN and GRUs with two ensemble approaches, soft voting (directly combines predictions through averaging) and stacking (uses an additional layer of learning to optimally integrate individual model predictions into a final outcome), and an optimized Multi-Head Attention transformer model. Hyperparameters of the model like the model dimension, number of attention heads, and transformer layers to minimize prediction errors were optimized using the Optuna framework. Regarding results, the hybrid CNN-GRU with attention model achieved high accuracy (MAPE of 6.333%), while the soft voting ensemble model attained the lowest MAPE of 6.125% and the Multi-Head Attention model MAPE of 6.889%. These results underline the effectiveness of attention-based mechanisms in capturing complex price dynamics.

Feature selection can also lead to improvements in accuracy. Wavelet transform (WT) employing filters, decomposes the complex price time series to manage inherent fluctuations, capturing breakpoints and discontinuities. This structure is then followed by a robust feature selection based on mutual information (MI) and entropy measures, efficiently ranking inputs by relevance and redundancy as proposed in [29]. This two-step preprocessing notably enhances the quality and informational efficiency of the inputs fed into the deep learning module. The final forecasting model leverages a deep learning architecture with LSTM cells, structured with 2 layers consisting of 200 and 100 hidden units respectively. The model's effectiveness is validated using datasets from the PJM and Spanish electricity markets and compared with several other models. The hybrid approach significantly outperformed the alternative methods across various metrics.

Decomposition strategies are also explored in [30]. The authors propose a hybrid day-ahead forecaster tailored to markets with high renewable penetration, in which the raw hourly series of price, wind and solar generation, and load for the Danish DK1 zone are first denoised with Empirical Wavelet Transform (EWT) to curb autocorrelation and non-stationarity. Each decomposed component is then passed to a three-layer LSTM enhanced with a feature level attention mechanism that dynamically weights the relative importance of the four exogenous inputs. Finally a Criss-Cross Optimization (CSO) algorithm fine-tunes the weights and biases. This EWT-AM-LSTM-CSO architecture outperforms both single-series variants (LSTM

without decomposition) and an array of EWT-driven machine learning baselines demonstrating that the combined use of EWT decomposition, attention-guided sequence modelling and CSO-based parameter improve accuracy.

A different two-step, decomposition-based deep-learning framework, denoted as VMD-CLSTM-VMD-ERCRF, is presented at [31]. In the first stage, the original Queensland spot-price series is variational-mode-decomposed (VMD) into up to 19 intrinsic mode functions whose optimal number is selected via the central-frequency-ratio criterion. Each sub-series is then normalized and modeled with a hybrid Convolutional LSTM (CLSTM) that couples CNN feature extraction with LSTM temporal gating. In the second stage, residuals from the CLSTM are themselves decomposed with VMD and the resulting error components are predicted with a Bayesian-optimized Random Forest, whose outputs are added back to the preliminary forecast to compensate systematic bias. Hyperparameters of both Random Forest and CLSTM are tuned using Gaussian-process Bayesian optimization with an adaptive learning rate. Seasonal and annual experiments show the proposed architecture outperforms a single-stage VMD-CLSTM and four other used benchmarks like plain LSTM and Random Forest.

2.4 Comparison and this Thesis

All studies that were presented in this section are summarized in Table 1 to make the comparison easier. The reviewed evidence suggests that no single method dominates across all settings. Each model is tailored at specific markets, thus their reported performance tends to be contingent on contextual factors such as data frequency, forecast horizon, forecast strategy and combination of predictors and historical values that were utilized to train the models. With that in mind, in this thesis the goal is to develop, test and compare methods from the previously mentioned categories to forecast the electricity price of the Day-ahead market (DAM) in Greece. All the developed methods will utilize historical prices and other predictor variables like RES generation, Load forecast and calendar dummies to predict 1-day ahead (24h) of prices. All methods will be compared in the same time period (2020-2024) and the comparison won't include only the ranking of the methods, but also how each additional predictor set, affects the final forecast.

Table 1: Summary of electricity-price forecasting studies

Authors/Year	Dataset	Methods	Predictor Variables	Forecast Horizon	Forecast Strategy	Best Method
Abunofal <i>et al.</i> [16] 2021	Germany, Case 1: 2015–2020, Case 2: 2016–2017	ARIMA, SARIMA, SARIMAX, MLR	Demand, renewable share, natural-gas price, hard-coal price, calendar dummies	1-day (24h)	Direct	SARIMAX
Carcia <i>et al.</i> [17] 2005	Spain (1999–2000) / California (Jan 2000–Dec 2000)	GARCH, ARIMA	Price lags, hourly demand	1-day (24h)	Recursive	GARCH + Demand
Yousefi <i>et al.</i> [18] 2019	California (2001–2017) monthly	SARIMA, logistic regression, SVM, KNN, RF	NG consumption, electricity from coke, net generation, imports, GDP, renewables split	3- & 5-year	Recursive(SARIMA)Logistic Reg. / Direct(ML)	
Plakas <i>et al.</i> [19] 2022	GR, BG, DE, AT (Jun 2021–May 2022)	RF & SARIMA for sub-modules	NG price, CO ₂ price, load fct, RES fct, wind speed, solar radiation, calendar, cross-border	1-day (24h)	Direct	–
Kızıldağ <i>et al.</i> [20] 2024	Turkey (Jan 2021–Sep 2023)	ARIMA, SARIMA, SVM, XGB, MLP, LSTM, Bi-LSTM, CLSTM	Calendar, macro-econ, RES split	1-hour	Direct	XGBoost
Poggi <i>et al.</i> [21] 2023	Germany (Jan 2020–mid-2022)	ARIMA, SARIMA, AR, XGB, LSTM	Price lags, calendar	24h / 168h / 720h / 6-mo daily avgs	Recursive(AR- type) / Direct(DL)	LSTM

Continued on next page

Table 1 (continued)

Authors/Year	Dataset	Methods	Predictor Variables	Forecast Horizon	Forecast Strategy	Best Method
Singhal <i>et al.</i> [23] 2011	N/A	3-layer ANN	Calendar, fcst demand, price lags	1-day (48 slots)	Direct	–
Uğurlu <i>et al.</i> [24] 2018	Turkey (Jan 2013–Dec 2016)	SARIMA, MS-AR, SETAR, 3-ANN, CNN, LSTM, GRU	Price lags, demand/supply bal., consumption, balancing price (24 h lag)	1-day (24h)	Direct	3-layer GRU
Tschora <i>et al.</i> [25] 2022	FR, DE, BE (2011–16 / 2016–21)	LEAR, RF, DNN, CNN, ChainSVR, MultiSVR	Price lags, calendar, load	prod. fcsts, RES, neigh. prices, gas price ...	1-day (24h)	Direct (LEAR, RF, ...) / Rec. (ChainSVR), FR: MultiSVR, DE:DNN, BE:ChainSVR
Lago <i>et al.</i> [26] 2018	Belgium (Jan 2010–Nov 2016)	DNN, LSTM-DNN, GRU-DNN, CNN	Price lags, load & gen. fcsts, FR prices	1-day (24h)	Direct	DNN
Kuo <i>et al.</i> [27] 2018	PJM (2017)	SVM, RF, MLP, CNN, LSTM, CNN-LSTM	Price lags	1-hour	Direct	CNN-LSTM
Laitsos <i>et al.</i> [28] 2024	Germany (Oct 2018–Aug 2023)	CNN(+attn), CNN-GRU(+attn), MH-Attn, CNN-GRU+Transf.	Price lags, fuel-specific gen. types	1-hour rolling	Direct	CNN-GRU+Transf.
Memarzadeh <i>et al.</i> [29] 2021	PJM (2006, 2018), Spain (2002)	WT-FS-LSTM vs benchmarks	Price lags	1-week (168 h)	Recursive	WT-FS-LSTM

Continued on next page

Table 1 (continued)

Authors/Year	Dataset	Methods	Predictor Variables	Forecast Horizon	Forecast Strategy	Best Method
Meng <i>et al.</i> [30] 2022	Denmark-DK1 (2018–2019)	EWT-AM-LSTM-CSO vs others	Price lags, wind & solar gen., load fct	24 × 1 h	Direct	EWT-AM-LSTM- CSO
Ghimire <i>et al.</i> [31] 2024	Queensland (2014–Oct 2022)	VMD-CLSTM-VMD- ERCRF vs others	Price lags	1-step (½ h)	Direct	VMD-CLSTM- VMD-ERCRF

3 Examined Forecasting Methods

In this section we present the methods used in this thesis to forecast the DAM price of Greece. Based on the insights drawn from the Literature Review, the forecasting methods examined in this thesis were selected due to their wide adoption and proven effectiveness in the field of Electricity Price Forecasting. SARIMA and SARIMAX represent well-established statistical approaches frequently employed to capture seasonality and exogenous influences. Random Forest and XGBoost have been consistently highlighted as robust machine learning models capable of handling nonlinearities and complex relationships. Finally, LSTM networks are among the most prominent deep learning architectures for sequential data, making them suitable for electricity price dynamics. Examining these representative methods not only allows for benchmarking against established practices but also provides a solid foundation for future investigations into hybrid and different forecasting approaches.

3.1 Autoregressive Integrated Moving Average

Statistical methods forecast the current price by using a mathematical combination of the previous prices and/or current values of exogenous factors. The most popular methods in this category are *additive* ones, meaning that the predicted price is the sum of components and several factors. Regression is one of the most widely used statistical techniques: the general purpose of multiple regression is to learn more about the relationships between several independent (predictor) variables and a dependent (criterion) variable. One of the most popular methods in this category for time-series forecasting is the **Autoregressive Moving Average (ARMA)** model, which comprises two sub-models: Autoregressive (AR) and Moving Average (MA).

An AR model of order p , denoted $AR(p)$, predicts the variable of interest using a linear combination of its past values. This model takes into account the random nature and time correlations of the phenomenon studied, starting with previous prices. Equation (1) describes the AR model with P_t being the target variable at time t , P_{t-i} , its i -th lagged value, $\phi_1, \dots, \phi_p$, the coefficients of the AR polynomial, and ε_t a term

that takes into account a randomness component, being modeled by a white noise stochastic process.

$$P_t = \alpha + \phi_1 P_{t-1} + \phi_2 P_{t-2} + \cdots + \phi_p P_{t-p} + \varepsilon_t. \quad (1)$$

Likewise, a MA model predicts a variable using the linear combination of past forecast errors. This model considers q previous values of the noise and is denoted as $\text{MA}(q)$. The respective equation (2) describes the model and variables are defined as above, with $\theta_1, \dots, \theta_q$ being the MA coefficients.

$$P_t = \alpha + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}, \quad (2)$$

If these two methods are combined, we get the **ARMA**(p, q) model where p and q are the coefficients of AR and MA, respectively. In the **ARMA**(p, q) model, the price is defined as in Equation (3).

$$\varphi(B)P_t = \theta(B)\varepsilon_t, \quad (3)$$

where B is the backward-shift operator ($BP_t = P_{t-1}$) and

$$\varphi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p, \quad \theta(B) = 1 + \theta_1 B + \cdots + \theta_q B^q.$$

$\phi_1, \dots, \phi_p$ and $\theta_1, \dots, \theta_q$ represent once more the coefficients of the AR and MA polynomials and ε_t a collection of independent and identically distributed Gaussian random variables, $\varepsilon_t \sim \mathcal{WN}(0, \sigma^2)$, hence specifying the white noise component cited before at each time of interest.

ARMA models are suitable only for *stationary* time series, i.e., time series whose properties do not depend on the time at which the series is observed: time series with trends, or with seasonality, are thus non-stationary. To handle non-stationary data we difference the series, leading to the **Autoregressive Integrated Moving Average (ARIMA)** model. Differencing in statistics is a transformation applied to a non-stationary time-series in order to make it trend-stationary (i.e., stationary in

the mean sense), by removing or subtracting the trend or non-constant mean. This is done with the introduction of the lag- d differencing operator defined in equation (4).

$$\nabla^d P_t = P_t - P_{t-d} = (1 - B^d)P_t. \quad (4)$$

With d such differencing passes, an **ARIMA**(p, d, q) process is described in equation (5)

$$\varphi(B) \nabla^d P_t = \theta(B) \varepsilon_t. \quad (5)$$

When the time series is also characterized by seasonality (i.e., repeating circular pattern), a standard approach is to use the **Seasonal Autoregressive Integrated Moving Average (SARIMA)**, an extension of the ARIMA model. Equation (6) defines the **SARIMA**(p, d, q) $x(P, D, Q)_s$ where (p, d, q) refers to the non-seasonal component, (P, D, Q) refers to the seasonal one, and s indicated the number of observations in a season. $\Phi(B^s)$ and $\Theta(B^s)$ are the seasonal polynomials and ∇_s^D the seasonal differencing.

$$\Phi(B^s) \varphi(B) \nabla^d \nabla_s^D P_t = \Theta(B^s) \theta(B) \varepsilon_t. \quad (6)$$

Finally, incorporating additional exogenous time series as predictors to the previous models, generalizes these univariate models to their multivariate counterparts: **ARMAX**, **ARIMAX**, and **SARIMAX**.

3.2 Random Forest

Random Forest is an ensemble learning technique that builds a large collection of decision-tree predictors and lets them vote to produce a single, robust prediction. Each individual tree is grown on a bootstrap sample of the training data and this “bagging” step injects diversity because every sample is drawn with replacement, so some observations are repeated while others are omitted. At every node of the tree only a random subset of the candidate predictors is considered for splitting, which further de-correlates the trees by ensuring that dominant variables do not appear in the same relative positions across all trees. Once a forest is grown, a classification

task is solved by majority voting among trees, whereas in regression each tree outputs a numerical and the forest prediction is their arithmetic mean. Because the ensemble averages away the variance of the individual trees while preserving their low bias, Random Forest often attains accuracy comparable to more complex non-linear models.

Adapting Random Forest to time-series forecasting requires reconciling two characteristics of temporal data that do not exist in independent and identically distributed settings: autocorrelation among observations and the unidirectional flow of time, which prohibits shuffling the data indiscriminately. The most widespread strategy is to re-frame the forecasting problem as a tabular regression task by transforming the univariate or multivariate series into a supervised-learning matrix. For a univariate series y_t , a sliding window ℓ is defined so that at each time stamp t the predictors are past values $y_{t-1}, y_{t-2}, \dots, y_{t-\ell}$ and the target is the future value y_t or y_{t+h} for an h -step-ahead forecast. When exogenous regressors are available, contemporaneous or lagged versions of these variables are concatenated to the same feature vector. This lag-embedding converts temporal dependence into cross-sectional co-variation that a tree can split on, allowing the forest to discover non-linear interactions among lags and between the series and external drives.

Although the core algorithm remains unchanged, some care is necessary when training and validating a Random Forest on temporally ordered data. Bootstrapping rows must respect the chronological structure, so practitioners typically sample entire contiguous blocks rather than individual observations. This block bootstrap preserves local serial correlation while still providing resampled diversity.

When configuring a Random Forest, the central hyper-parameters that need to be chosen, include the number of trees, the maximum depth, and the number of features tried at each split. More trees generally reduce variance at the cost of longer training and inference times. Within each tree, shallower, more heavily regularized architectures curb overfitting and deeper trees capture subtler interactions. The number of features parameter governs how many predictor variables are randomly sampled when considering a split, balancing tree decorrelation (small values) against individual tree strength (large values).

3.3 XGBoost

Extreme Gradient Boosting is a gradient-boosted decision-tree algorithm that refines the original boosting framework by engineering both algorithmic and systems-level improvements. Conceptually, gradient boosting fits an additive model

$$F_M(\mathbf{x}) = \sum_{i=1}^m \eta f_i(\mathbf{x}), \quad f_i \in \mathcal{F},$$

in which each leaf learner f_m is trained to predict the residual error, or more precisely the negative gradient, of the ensemble accumulated so far.

XGBoost distinguishes itself by using second-order Taylor expansions of the loss during tree construction, so each split is chosen with awareness of both first- and second- order gradient statistics, leading to faster convergence and more nuanced partitions. The method further incorporates regularization on leaf weights, a shrinkage factor that scales each new tree's contribution, and column-wise subsampling that injects diversity akin to Random Forest but without destroying boosting's sequential error-correcting mechanism.

Applying XGBoost to time-series forecasting follows the same overarching strategy used with other tree-based ensembles: the temporal sequence is first recast as a supervised learning problem by embedding lagged values and calendar or exogenous co-variates into a tabular design matrix. For a univariate series y_t a look-back window of length ℓ is defined and forms an input vector $y_{t-1}, y_{t-2}, \dots, y_{t-\ell}$ augmented by engineered features like rolling means, seasonal dummies and exogenous variables. Forecasting h -steps ahead can proceed in recursive fashion, feeding each point forecast back into the feature set, or in direct fashion, where a separate boosted ensemble is trained for each horizon.

Similarly to Random Forest the key hyperparameters that need to be chosen, involve the number of trees, the maximum depth of the trees and in this case of XGBoost we need to chose the learning rate, the regularization coefficients and the column/row subsample ratio to inject randomness. If hyperparameter tuning is applied, it must be embedded within a forward chaining framework to estimate true out-of-sample performance.

3.4 Long Short-Term Memory

Long Short-Term Memory (LSTM) networks belong to the family of recurrent neural networks (RNNs) and are engineered to specifically overcome the vanishing-gradient problem that hampers other RNNs when learning long-range dependencies, such as the Gated Recurrent Unit (GRU). Instead of a single recurrent transformation,

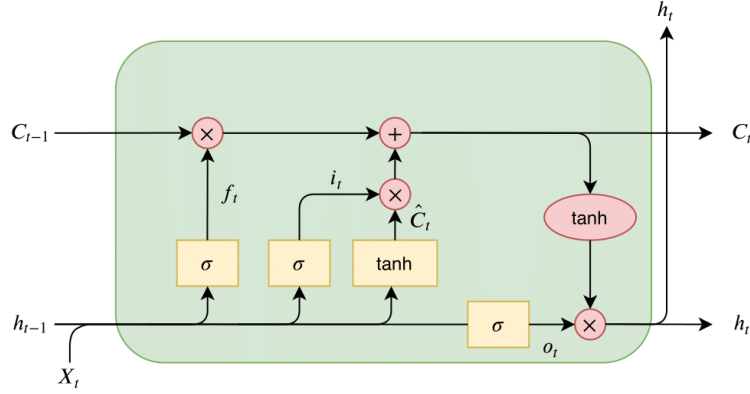


Figure 2: LSTM memory cell Architecture [32].

an LSTM cell maintains an internal memory state c_t that is modulated by three multiplicative gates (forget, input and output gates), which allow the network to decide what information to keep, discard, or expose at each time step. Denoting the input vector at time t by $\mathbf{x}_t$ and the previous hidden state by $\mathbf{h}_{t-1}$, the canonical Equations (7) describe the LSTM memory cell. Figure 2 also helps visualize the memory cell of an LSTM neural network.

$$\begin{aligned}
 f_t &= \sigma(W_f \mathbf{x}_t + U_f \mathbf{h}_{t-1} + b_f), \\
 i_t &= \sigma(W_i \mathbf{x}_t + U_i \mathbf{h}_{t-1} + b_i), \\
 \tilde{c}_t &= \tanh(W_c \mathbf{x}_t + U_c \mathbf{h}_{t-1} + b_c), \\
 c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\
 o_t &= \sigma(W_o \mathbf{x}_t + U_o \mathbf{h}_{t-1} + b_o), \\
 h_t &= o_t \odot \tanh(c_t).
 \end{aligned} \tag{7}$$

In those equations, $\sigma(\cdot)$ is the logistic sigmoid, a mathematical function whose graph has a characteristic S-shaped curve, being convex for values less than zero

and concave for values greater than zero. In addition, $\odot$ denotes element-wise multiplication and W_*, U_*, b_* are learned parameters. The forget gate f_t scales the previous memory, ensuring that irrelevant or outdated information decays. The input gate i_t moderates how much of the candidate update $\tilde{c}_t$ is written into memory, and the output gate o_t controls the exposure of the updated cell state through the hidden representation h_t . Because the gradient can flow directly through the additive link in the cell state, long-term dependencies propagate without being repeatedly multiplied by sub-unitary weights, allowing the network to learn patterns that span hundreds of steps.

In a time-series forecasting context, the raw sequence, whether a univariate sensor reading or a multivariate panel of indicators, is fed to the LSTM either one value at a time or in minibatches of fixed-length windows. The network ingests a look-back horizon of ℓ steps $x_{t-\ell+1}, \dots, x_t$ processes them sequentially through the recurrent layers, and emits a forecast y_{t+h} at the final step, where h is the prediction horizon of interest. Multi-step forecasting can proceed in one of three regimes. In the recursive mode a single-step LSTM prediction is iteratively rolled forward, using its own previous outputs as inputs. In the direct case a separate network is trained for each horizon and in “sequence-to-sequence” mode the encoder LSTM digests the historical window into a context vector while a decoder LSTM unfolds that context into an entire path in one forward pass, thereby avoiding error accumulation.

Several architectural choices arise when tailoring LSTMs to practical forecasting problems. Stacking multiple recurrent layers deepens the model’s representational power, allowing lower layers to capture short-term interactions while higher layers understand longer-term motifs. Selecting the amount of layers belongs to the task of hyperparameter tuning. The number of recurrent layers and the dimensionality of each hidden state determines how much hierarchical and long-range information the network can store balancing computational time and inflate memory usage. Complementary to depth and width is the look-back window, or sequence length, fed into the network, which controls how far into the past the model explicitly sees. In addition, optimizer settings govern convergence speed and stability, while batch size affects gradient variance and hardware utilization. Finally, training loop controls such as early-stopping patience and the number of training epochs interplay with all the above to decide when the model has extracted the salient temporal patterns without drifting into over-fitting.

4 Collection and Analysis of Data

4.1 Data Collection

To develop and train the forecasting models presented in this thesis, it was necessary to assemble a high-quality and complete dataset containing the time series of the Greek Day-ahead Market (DAM) clearing price, which is the target variable that all explored models aim to predict with as much accuracy as possible. Robust learning algorithms, as was discussed also in previous sections, require large volumes of chronologically ordered data that span multiple seasonal cycles so that weekly, annual and weather-driven patterns are all represented. Moreover, every observation of the target price must be accompanied by its potential predictors on exactly the same timestamp.

The most authoritative public source for pan-European electricity market information is the ENTSO-E Transparency Platform [33], operated by the European Network of TSOs in compliance with the respective European regulation. Besides price timeseries, the platform publishes forecasts and realized values for load, generation by technology, cross-border flows, among other system indicators. The platform exposes a RESTful API that returns machine-readable JSON/XML payloads, enabling fully automated data pipelines.

Specifically for this thesis the platform was accessed through the open-source Entsoe-pandas-client [34], which wraps the raw API and returns results directly as pandas series, which can then be processed easily using python. In the extraction script that was developed, some parameters need to be adjusted to each case to download data successfully. Firstly the user needs to hold an API key, which is a personal token issued by ENTSO-E after free registration, and accompanies each request to the system. In addition, the user needs to pick the time window he is interested in, parsed into timezone-aware Brussels-time timestamp objects. In our case the time window spans from 1st January 2018 to 31st December 2024. Additionally, the user needs to provide the specific bidding-zone code for the country he is looking to acquire the data. In our case the code is “GR”, which is the two-letter Energy Identification Code (EIC), for Greece. Finally the user needs to include the specific client methods

(queries) that specify which datasets to retrieve (e.g. DAM price, load forecast etc.).

The resulting CSV offers an hourly panel in which the Greek DAM price is flanked by three physically meaningful co-variates, system load forecast, solar-generation forecast, and wind-generation forecast, for the exact same timestamps. These extra variables are the main exogenous variables, among other engineered features, that expand the information well beyond price history alone and will be utilized to improve the accuracy of the developed models. Their preprocessing, feature engineering and statistical relevance will be addressed in the following sections.

4.2 Data Preprocessing

Every model developed in this thesis, expects the target variable and its exogenous regressors to be sampled at uniform one-hour intervals. Missing observations break this assumption, because the algorithms cannot infer appropriate lags or construct input windows of equal length. Conversely, duplicated timestamps overweight specific hours and distort the empirical autocorrelation structure. These abnormalities can be tangled also with the 24 hour horizon imposed to all the models. With that in mind, two kinds of irregularities had to be resolved.

The first issue was true data gaps, as minimal days of data lacked some predictors. For example, on 27 October of 2024 the ENTSO-E platform did not yet hold the Greek, load, solar and wind forecasts, so the corresponding cells were returned as *NaN*. Instead of dropping that day of data and reducing the dataset, these missing points were imputed with the midpoint of their temporal neighbors, the arithmetic mean of the same hour one day before and one day after.

The second complication arose from Daylight-Saving-Time (DST) shifts. Although the ENTSO-E API itself delivers records in Coordinated Universal Time (UTC), our downstream processing converted timestamps to local Greek time (Europe/Athens) so that all variables share the same clock as the wholesale market. During this conversion the spring and autumn DST transitions introduced calendar anomalies. Specifically when clocks advanced in March, the hour between 2:00 and 3:00 AM was skipped, leaving affected dates with only 23 rows instead of 24. Conversely, when clocks retreated in October, the hour between 2:00 and 3:00 was repeated, so

these dates contained 25 rows. The remedial strategy mirrored the logic used for true gaps. For the 23-hour days the missing 2:00 AM row was synthesized by averaging the adjoining 1:00 and 3:00 values of each series. For the 25-hour days the second occurrence of 2:00 AM was removed, retaining only the first instance so that no hour is represented twice.

Consequently, after these corrections the dataset formed an uninterrupted, perfectly regular hourly panel. This homogeneity is crucial as it allows matrix slicing operations to produce identically shaped input tensors and keep autocorrelation diagnosis interpretable.

4.3 Feature Selection and Data Analysis

Electricity price, and by extension forecasting of the price, sits at the intersection of power system engineering, economics, meteorology, politics and human behavior in general. Consequently, a first careful exploration of the raw data is indispensable. Figure 3 plots the entire hourly Greek Day-Ahead Market price from 2018 through 2024, together with the corresponding daily mean. Two high-level observations stand out immediately. Until mid 2021, clearing prices hovered between 40 €/MWh and 70 €/MWh, interrupted only by occasional spikes linked to local fuel shortages or sudden outages. From the second half of 2021 the series embarks on a steep ascent that culminates in extreme hourly prices and daily averages above 400 €/MWh. This regime shift mirrors the broader European energy crisis, which started with the post COVID-19 demand rebound, was magnified with tight Liquefied Natural Gas (LNG), and peaked after Russia's invasion of Ukraine in February 2022. The blue hourly prices around the daily mean, on the plot, reveal a forest of needle-like excursions who place single hours priced five to ten times higher than the surrounding baseline. Such spikes often arise from a confluence of scarce flexible generation, binding interconnector limits and abrupt renewable forecast errors. Their presence foreshadows the statistical challenges that the developed models will face in the forecasting process, as any model must cope with the occasional jumps that dominate error metrics.

To further examine the evolution of prices throughout the years, Figure 4 decomposes the series into monthly averages for each calendar year. The pre-crisis period (2018-

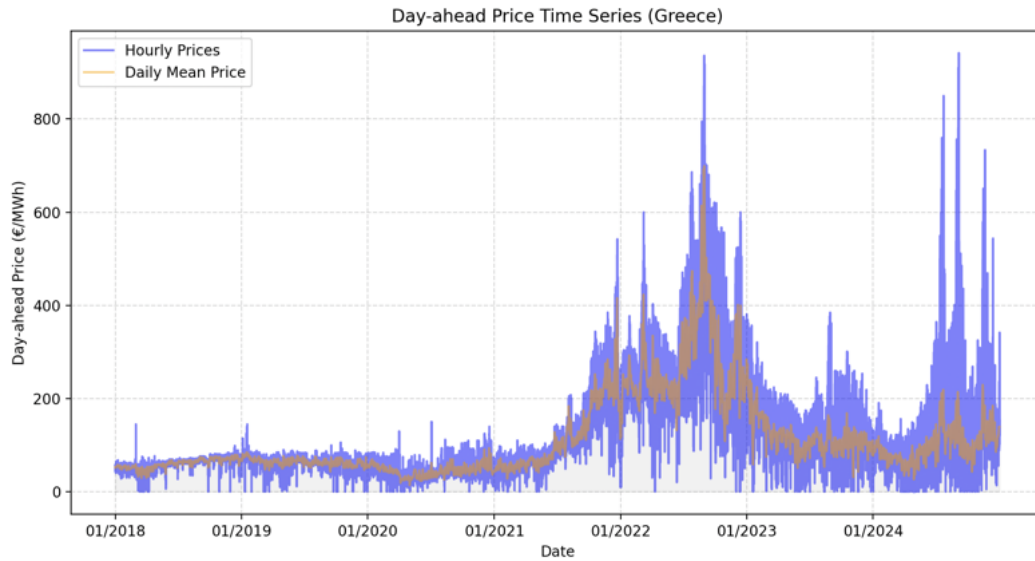


Figure 3: Greek DAM price timeseries over daily mean (2018-2024).

2020) exhibits the usual pattern of Mediterranean countries, where lower prices appear on months like April, May, and October when mild weather depresses both heating and cooling demand, and slightly higher levels in winter and the tourist driven summer peak. The pandemic year 2020 even shows a dip below 30 €/MWh in April, reflecting the severe lockdown. The lines related to 2021 and especially 2022 break this symmetry. From July 2021 onwards every month is costlier than the historical maximum and by August 2022 the average exceeds 430 €/MWh noting nearly 800% increase over the same month in 2019. The subsequent retreat during 2023 traces the easing of gas prices after Europe's emergency storage campaign, yet the curve never returns to pre-crisis levels with the price riding an upward trend in mid-2024 and reaching a new stable point at the end of the year.

Closer inspection on Figure 3 reveals several hours where the DAM cleared at negative values (down to -15€/MWh). Such events occur when there is an excess of production compared to consumption. Negative prices appear with a high number of renewables feed-in, low demand (e.g. weekends or holidays), and a high inflexible power generation. In some cases, inflexible power sources like lignite or nuclear are effectively willing to pay to remain online because shutting down and re-starting would be even more expensive or technically infeasible. Therefore, negative price episodes signal surplus variable renewable energy and limited storage or interconnection capacity.

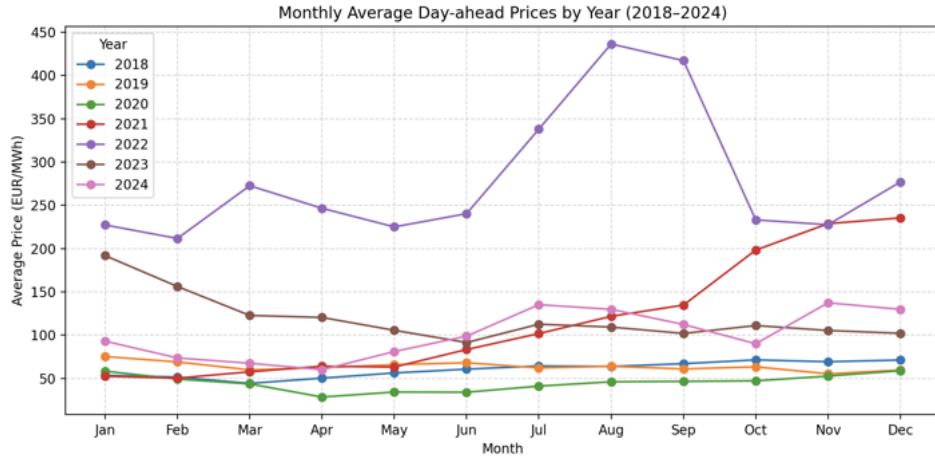
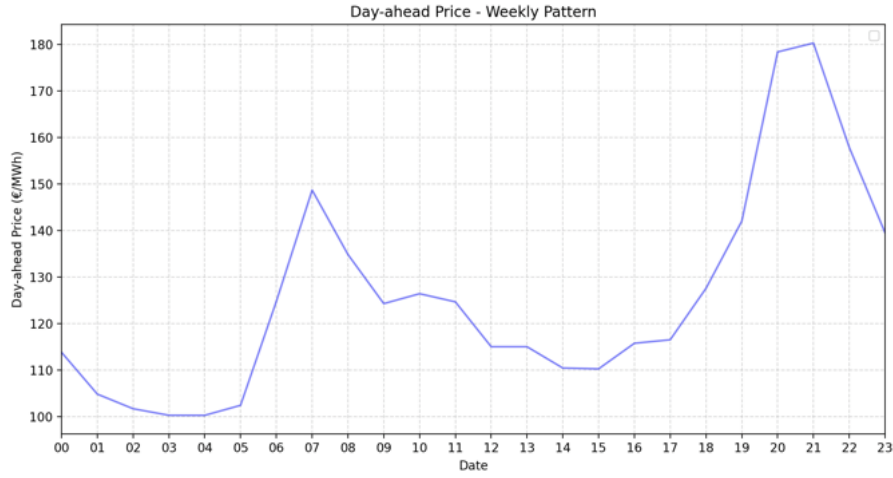
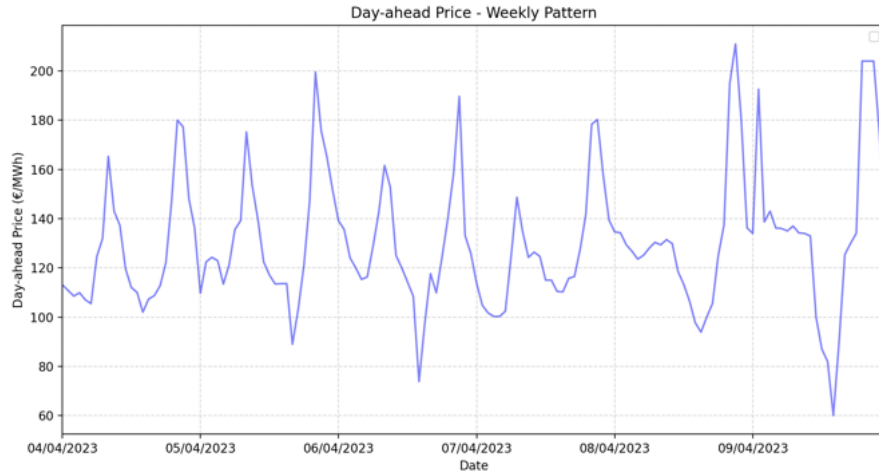


Figure 4: Monthly average DAM price by year (2018-2024).

All these extreme spikes and unstable behavior that the previous analysis indicates, makes the problem of price forecasting seem very difficult. Although this observation is quite accurate, based on the turmoil of recent years, electricity and as a matter-of-fact prices are still generally governed by rhythm and specific trends. People wake up, factories schedule shifts, and the sun rises and sets with deterministic frequency. By examining further, the behavior of electricity prices clear patterns and specific behavior reveals. Figure 5a shows the daily cycle of the prices. Specifically, price is the lowest from midnight until the early-morning hours when the demand is the lowest. Then the price climbs as the day progresses and the electricity needs increase. The price experiences another deep at the middle of the day around 12:00 – 16:00 when the RES penetration maximizes. Finally, the price peaks around 19:00 – 22:00 when the demand increases and the PV generation decreases drastically in comparison to the previous hours. If we also observe the weekly patterns in Figure 5b we can see that the daily cycle, we addressed earlier, indeed repeats on a day-to-day basis. In general, average prices on the weekends are slightly lower, mirroring industrial demand schedules. This difference is smaller during the summer period when the tourism boosts weekend consumption.



(a)



(b)

Figure 5: DAM Price Daily (a) and Weekly (b) patterns.

This observed weekly pattern, visually indicates that the price recorded at a given hour is highly correlated with the price in the same hour on preceding days or even, correlated with the prices the exact previous hours. Figure 6 formalizes this intuition. This Figures show the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF). ACF measures the total linear correlation between the series and a lagged copy of itself at $lag - k$. Tall stems, imply a statistically significant correlation at that lag. Similarly, PACF measures the direct correlation between the series and its $lag - k$ value after removing the influence of all shorter

lags. A significant stem means that the particular lag contributes unique information beyond earlier lags.

Specifically, the ACF on the examined dataset, forms a gentle wave with crests roughly every 24 lags (24h, 48h, 72h...). This indicates that the market tends to repeat its behavior day-by-day. A slight rise is also visible near lag 168 (24h x 7), hinting that weekdays share similarities beyond the daily rhythm, like weekly seasonality. The PACF indicates that the most powerful direct lag of the current price is the price one hour earlier. A second group of significant bars appears at multiples of 24 hours mirroring the daily pattern detected in the ACF.

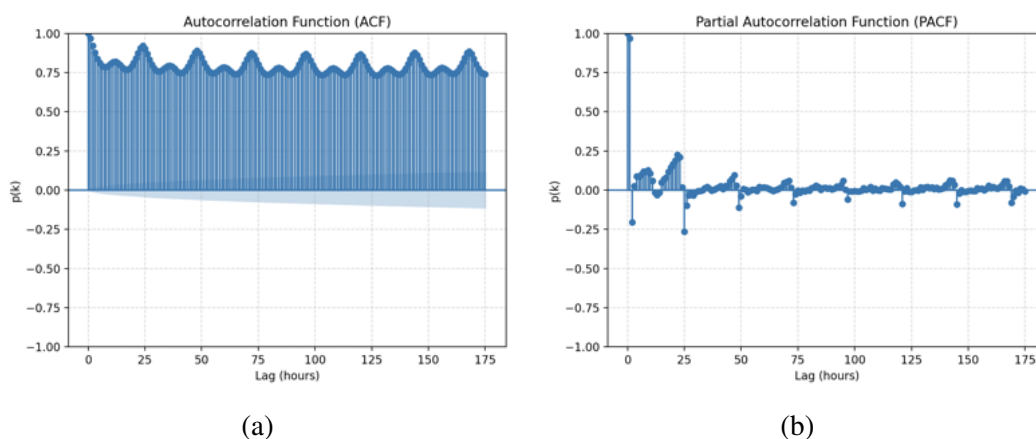


Figure 6: ACF (a) and PACF (b) plots.

Figure 7 takes the idea of daily correlation of prices one step further. Each cell of the heatmap, contains the Pearson coefficient between hour h on day D and hour h on day $D-1$. Correlations above 0.9 along the main diagonal mean that yesterday's 9:00 price, for example, is an excellent predictor of today's 09:00. High values above 0.9 are also found in the "nearby" hours emphasizing once more the correlation of the first lags, for example hour 0:00 of day D is highly correlated with hour 23 of day D as they are only one step apart. All these insights that were addressed in this section guide the selection of lagged inputs in both statistical and machine/deep learning models that were developed.

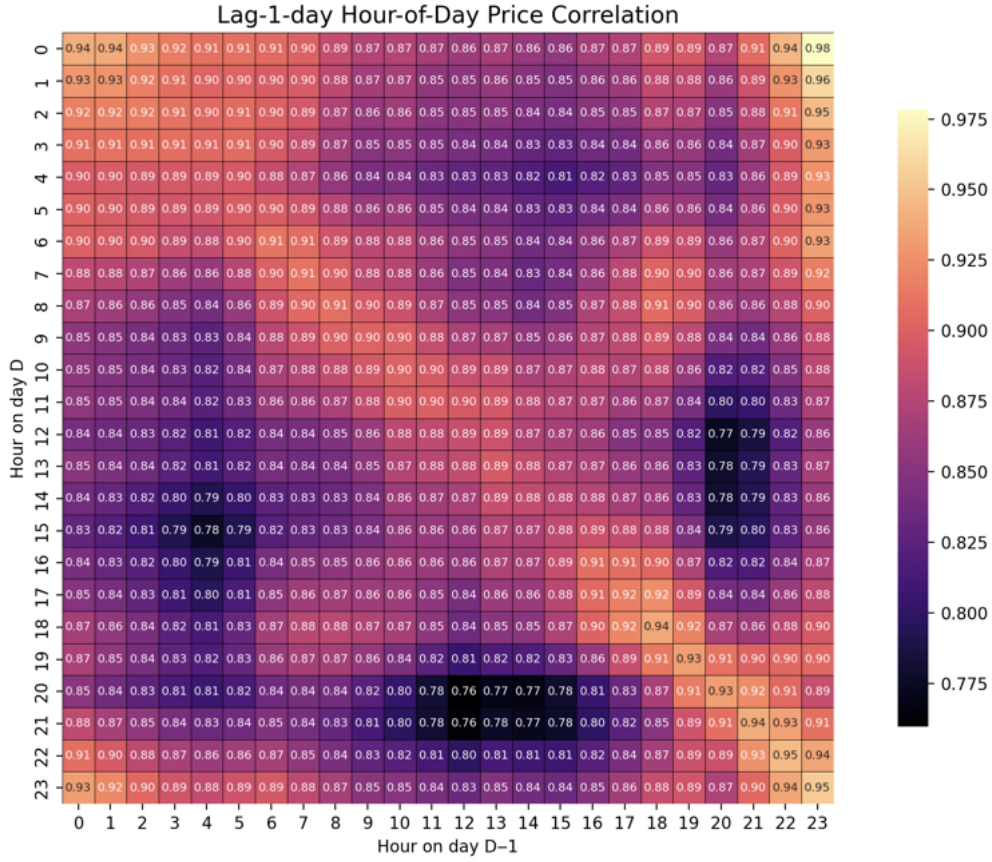


Figure 7: Heatmap of the correlation coefficients of electricity prices from the previous day to the next day.

Purely autoregressive specifications can leverage short-range inertia but remain blind to the physical and behavioral factors that cause prices to move. To supply this external signal, as was presented earlier in the data collection section, three series that are published at the day-ahead horizon are appended to the dataset. Specifically, system-load forecast, solar-generation forecast, and on-shore wind-generation forecast. Figure 8 visualizes the same representative weak, as Figure 5b for prices, of these forecasts. The blue load curve traces a pronounced weekday rhythm with a morning ramp at 8:00 and evening peak around 21:00, a repeated weekly pattern. Similarly, the orange solar band exhibits a textbook bell shape that repeats almost perfectly from one day to the next, peaking after solar noon and collapsing to zero at night. Obviously, the range of peaks may differ through the deferent seasons and is affected by cloudy days, but the general structure remains the same. The green wind series meanders more irregularity but still provides valuable information about the total day-ahead RES generation.

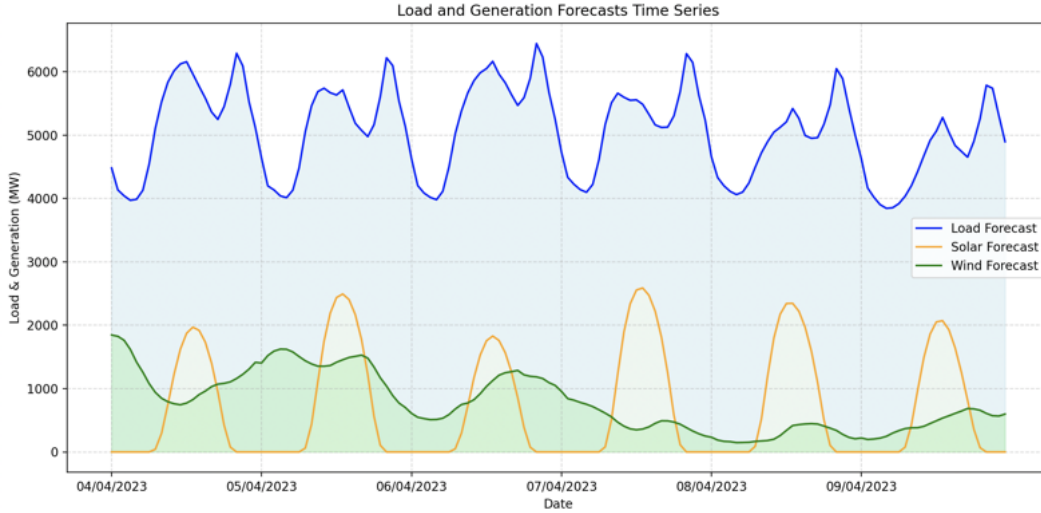


Figure 8: Load and Generation Forecasts, Weekly pattern.

These patterns matter because they explain much of the diurnal and variation observed in the price series. For example, the midday price dip noted earlier coincides with the crest of the solar curve, while the steeper evening ramp is amplified whenever wind speeds subside. Incorporating the forecasts therefore can give the models actionable information and not let them infer supply-demand balance solely from past prices.

4.4 Dataset Preparation and Splitting

Alongside the meteorological drivers the final dataset that was created, also consists compact deterministic calendar descriptors that capture the recurrent structure documented in the ACF/PACF and heat-map analysis. These are essentially engineered features that are noise-free, and they allow both statistical and machine learning methods generalize across years and capture daily/weekly seasonal patterns. Their inclusion completes a feature matrix that is simultaneously parsimonious and expressive. The final dataset created spans from 01/01/2018 – 31/12/2024 and each timestamp consists of the following fields:

- **Day-ahead Market Price**
- **Day-ahead total Load Forecast**

- **Day-ahead Solar Generation Forecast**
- **Day-ahead Wind Generation Forecast**
- **Hour timestamp:** takes values 0.0 – 23.0 for each 24 hours of the day
- **Weekday dummy:** takes values 0.0 – 6.0 indicating day of the week
- **Business dummy:** takes value 1.0 if the weekday is a business day, 0.0 otherwise

Before model fitting, an important step is to allocate observations into training and testing subsets. Training set is used for the model to learn the patterns and dynamics of the time series and then fit the appropriate parameters for each model. Test set on the other hand, is a data set used to provide an unbiased evaluation of the final model to examine its accuracy in the predictions it outputs on unseen data. In time-series forecasting this operation, must respect chronology as data can only be used to predict the future and never the inverse. Random shuffling, which is a common strategy in other machine learning tasks is not applicable here as this would contaminate the evaluation with information that is unavailable in real time.

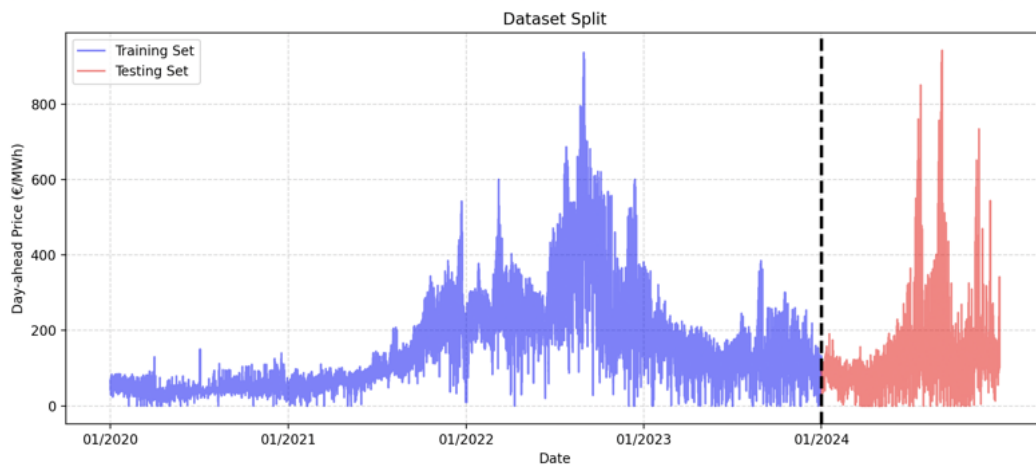


Figure 9: Splitting of the time series into training and testing sets.

Although the raw dataset starts on 1 January 2018, only the observations from 1 January 2020 to 31 December 2023 were retained for parameter estimation as experiments indicated that the first two years of data didn't align with the new price dynamics after 2020 and including them left the model accuracy unchanged while

increasing the computation time. Accordingly, all models were evaluated on the entire calendar year 2024, making an 80-20 ratio slice to the data. The exact splits are also shown in Figure 9. Using a complete year for testing is valuable as it exposes the models to the whole spectrum of meteorological and socioeconomic conditions they will encounter in live operation, while the results give complete context for model evaluation and comparison.

5 Experimental Setup and Implementation

5.1 Model Implementation Details

5.1.1 ARIMA setup

As discussed in Section 3, classical ARMA models presuppose stationarity. To verify this condition, we first applied the Augmented Dickey-Fuller (ADF) test. At the 5% significance level the ADF statistic rejected the null hypothesis of a unit root, ostensibly classifying the series as stationary. Visual inspection of the raw series (Figure 3), however, suggested the presence of a trend component, prompting a complementary Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test. In contrast to the ADF result, the KPSS statistic rejected stationarity at the same confidence level, signaling that deterministic or stochastic trends were indeed present. The combination of these results is indicative of a non-stationary time series. To eliminate the inferred unit root the time series was differenced once ($d = 1$) the differences trajectory in Figure 10 shows a constant mean, although the variance still exhibits mild heteroskedasticity.

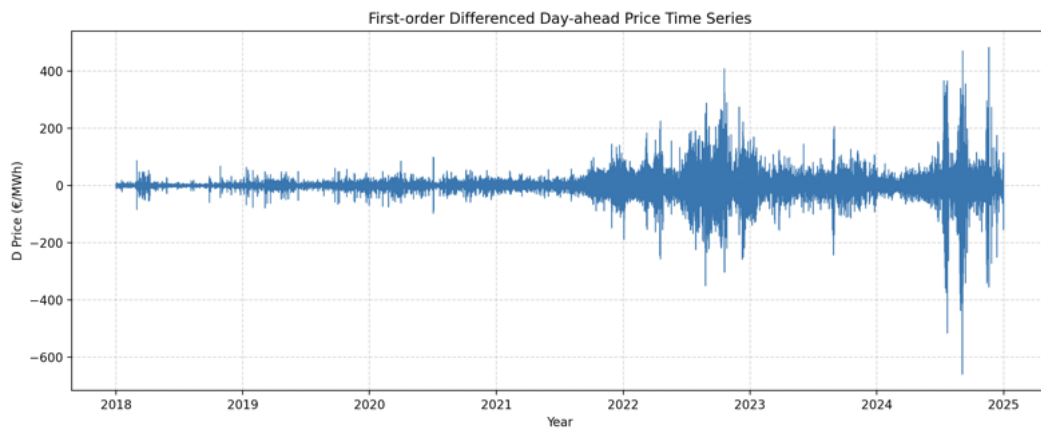


Figure 10: First-order Differenced Day-ahead Price Time Series.

Exploratory analysis and pattern recognition in Section 4 highlighted a strong weekly and daily seasonality. For that reason, a multiplicative $\text{SARIMA}(p, d, q) \times (P, D, Q)_s$ framework with $s=24$ was pursued. Because one non-seasonal difference already removed the long-term trend, seasonal differencing was unnecessary ($D=0$). Consulting with the corresponding Figure 11 of the ACF and PACF plots of the differenced time series the $\text{SARIMA}(1, 1, 1) \times (1, 0, 1)_{24}$ was the order selected.

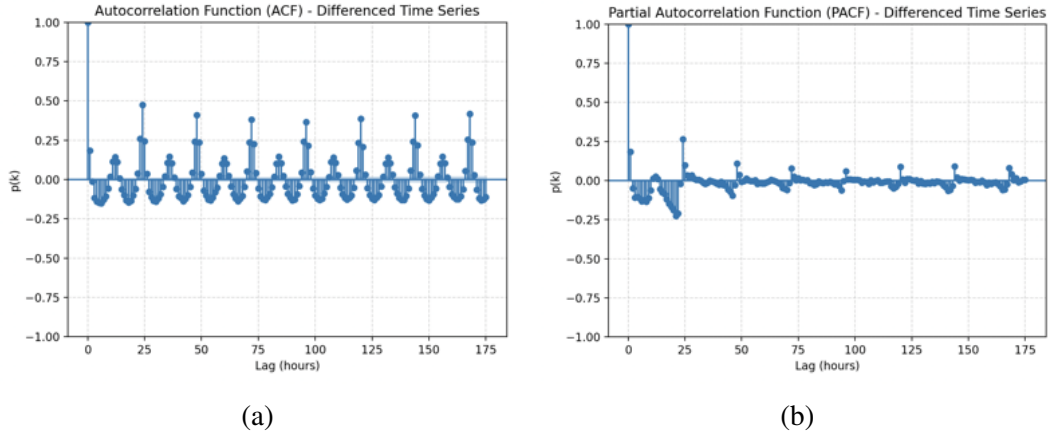


Figure 11: Differenced ACF (a) and Differenced PACF (b) plots.

This order selection also minimized the Akaike Information Criterion (AIC), which ranks competing statistical models by balancing explanatory power and simplicity by penalizing unnecessary complexity. Expanding the seasonal orders, for example to $(7, 0, 7)_{24}$, not only failed to reduce information criteria but also pushed the estimation times into multi-hour regime that made rendering these models impractical for iterative experimentation.

To emulate the operational workflow of an electricity market operator, the developed script for the SARIMA model, fits the baseline model on historical data up to 31 December 2023, and then the algorithm proceeds through the 2024 test set in 24-hour blocks. At each step the model predicts the next 24-hours and then updates the model with realized (actual) observations. This extends the state-space representation without a full re-estimation, so parameter values evolve only through the Kalman filter update rather than costly maximum-likelihood refit. The sliding-window, recursive strategy that was used, helps the model adapt to the newest market conditions in near-real time, while preserving computational efficiency.

5.1.2 Random Forest setup

The Random Forest method was deployed in an expanding-window, rolling-origin identical to the one-day-ahead protocol used for SARIMA. Because decision trees are invariant monotonic transformations, no scaling was required and instead the focus was to create a parsimonious yet information-rich feature set to adapt to the supervised-learning matrix. To help the model capture the patterns and seasonal dynamics that the price times series experiences, certain autoregressive lags were picked. Lag 1 captures the very strong hour-to-hour persistence. Lag 24 and lag 48 encode the diurnal rhythm and its carry-over to the next day and finally lag 168 (24×7) embeds the weekly cycle observed in previous sections. Experimental test with longer lags yielded negligible accuracy gains and higher memory cost, so the four-lag subset was retained. In addition to these lags the extra features of the dataset (exogenous forecasts and calendar descriptions) were utilized to improve the forecasting accuracy of the original model.

As far as the hyper-parameter selection goes, a coarse grid search showed that the error stabilized as soon as the forest contained about ninety trees. Consequently, the number of estimators was fixed at 100, a choice that ensured both statistical stability and fast retraining in the daily rolling schedule. The maximum depth of each tree was selected as 10, which preserved the forest's ability to model non-linear interactions yet guarded against over-fitting.

For every calendar day in 2024 the forest is refit from scratch on all data available strictly before that day, then used to predict the next day's 24 prices. Re-estimation rather than incremental updating is computationally feasible because trees train fast and allow each day's model to exploit the richer feature-response mapping induced by the expanding window.

5.1.3 XGBoost setup

Similarly to Random Forest, XGBoost was deployed in an expanding-window, rolling-origin. In addition the same feature set was selected to help the model adapt to the time series structure and create the supervised-learning matrix. In particular, the same exact autoregressive lags were picked. Lag 1 captures the very strong hour-to-hour persistence. Lag 24 and lag 48 encode the diurnal rhythm and its

carry-over to the next day and finally lag 168 (24×7) embeds the weekly cycle observed in previous sections. These features are once again complimented by the exogenous forecasts and calendar description dummies to increase the performance and accuracy of the model. Having the same feature set also helps with the accurate comparison of the results obtained from each method.

Hyper-parameter tuning for the XGBoost model included experimental search for the optimal parameter combination. The exercise revealed that predictive accuracy plateaued around 450 boosting iterations, while training time continued to rise almost linearly. Therefore, a number of 500 trees was selected. The maximum depth was again capped at 10, which is deep enough to capture high-order interactions, yet shallow enough to prevent the model of overfitting.

In contrast to Random Forest that grows many full-depth trees and relies on variance reduction through independent averaging, XGBoost grows trees sequentially and relies on iterative error-correcting updates. That's why it is important to guide how big each update is by picking a learning rate, and what notion of error it is trying to reduce, by choosing an objective. In our case a learning rate of 0.05 emerged as the optimal compromise and the squared-error objective was retained in its canonical form to align the optimization target with the Root Mean Squared Error metric used in model evaluation.

XGBoost follows the same forecasting protocol. At each test day the model is refit from scratch on the expanding window comprising every observation strictly earlier than the "current" day thus incorporating the newest market dynamics while preserving the long horizon memory. The freshly trained booster then produces a 24 hour forecast for next day, after which the true prices are revealed and appended to the training set, ready for the next cycle.

5.1.4 LSTM setup

LSTMs, and neural networks in general, require past context to make accurate forecasts. They learn patterns form sequences of past data and not from individual points. This is essential for the model to learn temporal dependencies, such as how past prices evolve over hours/days and how past weather and load forecasts affect future prices. For that matter, this mechanism expects the input to be a three-

dimensional tensor (batch x time-steps x features). An appropriate sliding-window generator builds samples (X,y) such that X represent the values up to the “current day” and y the values of the variable we are looking to predict. In our case 168 hours (7 days) was chosen as the input horizon based on the weekly cycle that was observed in the dataset analysis and a 24-h stride moves the window forward exactly one auction cycle as we are looking to have a one day-ahead horizon forecast. The sequence construction procedure can be visualized with the help of Figure 12. An

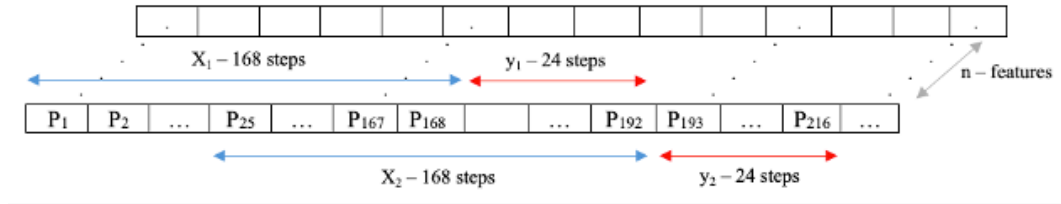


Figure 12: Preparation of Data sequences for LSTM.

important step before feeding data into a neural network is normalizing the input, as this improves convergence and training stability by ensuring consistent scale reducing the impact of different feature magnitudes. After the forecasting procedure the values are rescaled back to the original size to be examined.

As far as hyperparameter tuning for LSTM, it can be a difficult task as the parameters that need to be adjusted are various and they may affect each other. The final architecture of the designed LSTM that was selected, is the following:

- **LSTM layer with 64 units**, that captures medium-range temporal patterns without over-parameterizing
- **Dropout layer of 20%**, that mitigates co-adaptation of hidden units and combats over-fitting to price spikes
- **Batch Normalization Layer**, that stabilizes gradient flow and speeds up convergence
- **Dense layer of 24 neurons**, that produces the 24 hourly price forecasts simultaneously.

Stacking multiple LSTM layers with the same number of units was also examined, but that didn't translate into an improvement in performance, but an increase in computational time, so the simpler architecture was retained.

To compile the model, the Mean Squared Logarithmic Error was used that handles better the large outliers of the dataset. As for the optimizer, Adam was selected as the most common and popular one. The learning rate was picked $\eta = 0.001$ as it yielded the fastest reduction of validation loss during a coarse sweep ($\eta \in \{0.1, 0.01, 0.001, 0.0005\}$). As for training, a batch size of 32 balanced stochasticity and GPU utilization. The training run for 200 epochs but was guarded by early stopping (patience = 10) that stopped the training earlier when the validation loss stabilized and didn't improve further, preventing that way overfitting.

After the network was pre-trained on all data up to 31 December 2023, a lightweight recursive scheme kept it up-to-date during the 2024 evaluation period. Unlike tree-based models that we examined earlier, full retraining of the model after each 24 hour prediction proved infeasible due to high computational times. For this reason a different approach was used to mimic real world use. A clone of the pre-trained model is created each day with a lower learning rate (1×10^{-4}). In addition only the most recent 14 days of actual observations are used for adaptation, ensuring that the model reacts quickly to new information without forgetting long-term dynamics. The model is fine-tuned for 5 epochs with a small batch-size of 4. The adjusted network outputs the next 24 prices, which are stored and the prices are appended to the rolling-window, to repeat the cycle for the following day.

5.2 Evaluation Metrics

To be able to evaluate and compare the models developed, certain evaluation metrics need to be used. Rigorous comparison of forecasting models requires more than a single accuracy figure because each metric captures a different aspect of error behavior. Some of these metrics are also used as loss functions during the training processes of the machine learning models. We therefore picked Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Symmetric Mean Absolute Percentage Error (sMAPE) and Normalized MAE (nMAE). The equations of the errors are the following:

$$\text{MAE} = \frac{1}{24N_d} \sum_{d=1}^{N_d} \sum_{h=1}^{24} |p_{d,h} - \hat{p}_{d,h}|, \quad (8)$$

$$\text{RMSE} = \sqrt{\frac{1}{24N_d} \sum_{d=1}^{N_d} \sum_{h=1}^{24} (p_{d,h} - \hat{p}_{d,h})^2}, \quad (9)$$

$$\text{sMAPE} = \frac{1}{24N_d} \sum_{d=1}^{N_d} \sum_{h=1}^{24} \frac{|p_{d,h} - \hat{p}_{d,h}|}{|p_{d,h}| + |\hat{p}_{d,h}|}, \quad (10)$$

$$\text{nMAE} = \frac{\text{MAE}}{\frac{1}{24N_d} \sum_{d=1}^{N_d} \sum_{h=1}^{24} p_{d,h}} \quad (11)$$

where $p_{d,h}$ is the real price on day d and hour h , $\hat{p}_{d,h}$ the forecasted price on day d and hour h and N_d the number of days in the test dataset.

MAE (8) measures the average absolute error between predicted and actual values. Unlike RMSE it treats all errors equally and it gives a straightforward interpretation: how much the error deviates by €/MWh. RMSE (9), as mentioned, penalizes larger errors more as it squares the residuals. It helps assess how well the model handles large spikes or extreme price fluctuations. By extension sMAPE (10) is a normalized error metric that expresses the forecast error as a percentage. Such metrics allow comparisons across different time periods or markets. Since electricity prices fluctuate, a relative error measure is useful for understanding model performance in different price ranges. Finally, nMAE (11) normalizes the absolute error by dividing by the mean actual price. This makes MAE scale independent, allowing for comparisons across different datasets or markets.

6 Results and Analysis

This chapter documents the empirical performance of every forecasting technique developed in this thesis and discusses their comparative strengths. For each method we first report the accuracy of its univariate specification (trained solely on historical price data). We then successively enrich the model with the different predictor sets that are included in the dataset to examine how each addition affects the error metrics, with all results being benchmarked against the Persistence method that is also introduced. After identifying for every method the predictor set that yields the best performance, we put those “champion” configurations against each other on the common 2024 test horizon and the comparison focuses on accuracy and robustness across seasons.

All model development and experimentation were carried out in Python, leveraging well-established open-source libraries: Statsmodels for the SARIMA implementations and diagnostic statistics, scikit-learn for the tree-based ensembles and common preprocessing utilities, and TensorFlow for constructing, training and serving the LSTM networks. Supporting data-handling and visualization tasks relied on NumPy, pandas and Matplotlib.

Because operational deployment often hinges not only on accuracy but also on speed, the elapsed time required to train each model and produce the one-day-ahead forecast set is recorded and discussed alongside the error figures. Classical and tree-based models (SARIMA, Random Forest, XGBoost) were run on a macOS device equipped with a 2 GHz quad-core Intel Core i5 CPU, Intel Iris Plus Graphics 1536 MB GPU and 16 GB of RAM. Deep learning models (LSTM), require repeated large-scale matrix multiplications and back-propagation through time, which are markedly accelerated by parallel GPU architectures. To ensure that training and inference times remain comparable to the classical methods, the LSTM experiments were executed on Google Colab Pro using NVIDIA A100 GPU.

6.1 Model Results

6.1.1 Persistence Method (Benchmark)

In forecasting practice every new model is evaluated against a simple transparent alternative, often called Persistence model or Naïve Benchmark model. Without such a model it is impossible to say whether the additional complexity, data requirements and computational effort of more sophisticated techniques, such as statistical and machine learning, deliver a material improvement. A well-chosen benchmark therefore provides a sanity-check that more elaborate models clear minimum performance hurdle and an intuitive baseline that helps non-experts interpret subsequent accuracy figures.

The Persistence benchmark that was adopted in this thesis follows the common philosophy that future price is assumed to behave exactly like the recent past, with no explicit modeling of exogenous drivers. Concretely, we exploit the strong weekly seasonality of the series by averaging only the same type days in a short, fixed window [35]:

- **Monday - Friday:**

$$\hat{p}_{t,h} = \frac{1}{7} \sum_{i=1}^7 p_{t-i \cdot \text{bd},h}$$

where $p_{t-i \cdot \text{bd},h}$ denotes the same hour h on each of the previous seven business days.

- **Saturday - Sunday:**

$$\hat{p}_{t,h} = \frac{1}{7} \sum_{i=1}^7 p_{t-i \cdot \text{wd},h}$$

where $p_{t-i \cdot \text{wd},h}$ denotes the corresponding hour of the last seven weekend days (i.e. 3,5 weekends).

The rolling seven-day window is intentionally kept minimal so the benchmark remains a pure persistence rule that is quick to compute, avoids parameter estimation and uses only information that would have been available at the time of forecasting. A full 24-step horizon is produced for every day in the 2024 test set and the results

can be found on Table 2, along with the daily average of these metrics on Figure 13 that will both act as benchmarks for the more complicated methods that follow.

Model	MAE	RMSE	sMAPE	nMAE	Time
Persistence	21.860	40.860	27.423	0.217	–

Table 2: Performance metrics for the Persistence method (Benchmark).

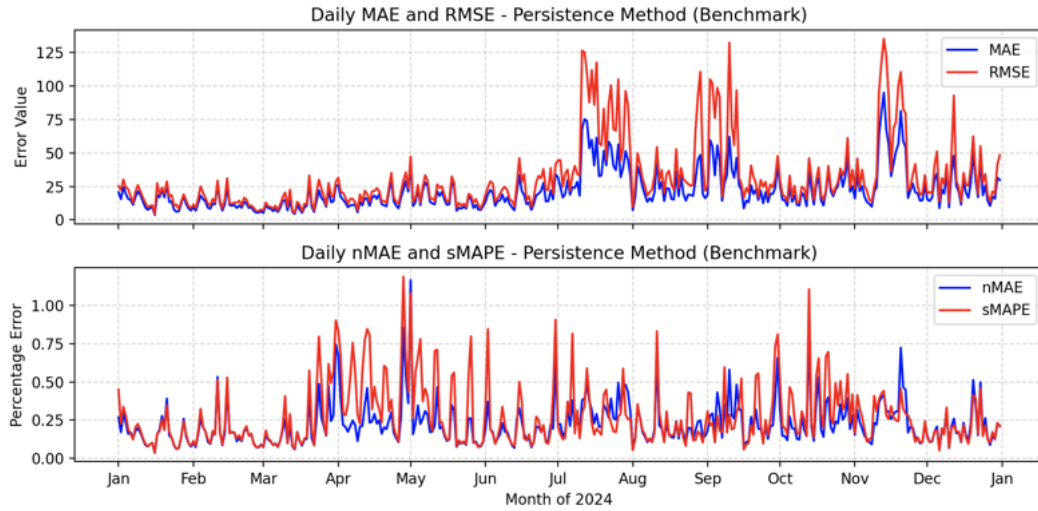


Figure 13: Evaluation Metrics Daily average - Persistence Method (Benchmark).

6.1.2 SARIMA Results

The results for every predictor set of the SARIMA method are found on Table 3. As the table indicates, the results obtained from the univariate model are worse than the ones we got from the Persistence model. The ARIMA order that was selected captures only daily seasonality. Electricity prices though, as was discussed in previous sections, exhibit strong seasonal patterns influenced by weekly cycles as well. If we want to capture this pattern with a SARIMA model, the order of the seasonal part needs to be increased drastically, assuming we are dealing with hourly observations. This has a result of increased computational loads and lead to the model not fitting properly and the program to crash. To take into consideration the weekly cycle pattern of the electricity price on the DAM instead of increasing the seasonal order of the SARIMA, which leads to memory issues, we approximated seasonality using Fourier terms:

$$F_t = \sum_{k=1}^3 \left[a_k \cos\left(\frac{2\pi kt}{168}\right) + b_k \sin\left(\frac{2\pi kt}{168}\right) \right]$$

$s_w = 168$ (weekly seasonality in hours)

$K = 3$ (number of harmonics)

a_k, b_k = Fourier coefficients

Each harmonic pair (sin, cos) captures one wavelength of the 168-hour cycle, the first harmonic describes the broad weekly undulation, while higher orders add progressively finer detail. Using the calendar dummies of the dataset to capture weekly cycle recorded worse accuracy to the Fourier terms so the Fourier terms were retained. Returning to Table 3, the univariate SARIMA, as stated earlier, seats

Model	MAE	RMSE	sMAPE	nMAE	Time
Persistence	21.860	40.860	27.423	0.217	–
SARIMA					
(univariate)	23.077	40.742	27.823	0.229	~26 min
SARIMAX					
(Fourier Terms)	21.144	38.175	26.229	0.210	~32 min
SARIMAX					
(Fourier, Load, Solar, Wind)	21.670	36.681	28.982	0.215	~38 min
SARIMAX					
(all predictors)	21.722	36.744	29.099	0.215	~44 min

Table 3: Performance metrics for SARIMA models.

slightly behind the Persistence benchmark on all four metrics, confirming that an ARIMA structure alone cannot compete with the strong day-of-week persistence already captured by the benchmark. Fourier augmentation delivers an immediate boost as MAE falls by ~9% relative to the benchmark model and nMAE drops below 0.22 for the first time. The gain stems from a much cleaner representation of the weekly cycle than the purely stochastic differencing could provide. Adding RES and load forecasts mainly lower RMSE, indicating fewer large outliers, yet the average MAE inches back up. Exogenous forecasts help on extreme days but add estimation variance in typical days. Finally adding the calendar dummies of the dataset on top of the Fourier terms, results in many highly collinear regressors and no truly new information. On the other hand they act as noise absorbers in estimation and

leave accuracy virtually unchanged while increasing computation time. Although the load, solar and wind forecasts resulted in only a better RMSE, this configuration was picked as the “champion” SARIMA model since our later model comparison focuses a lot on the models’ performance on the most volatile periods that exhibit the most extreme spikes and lower RMSE indicates better performance on these periods.

Figure 14, reveals how these modeling choices translate into calendar-time performance on the 2024 test set. On winter and early spring, errors are low and stable with the SARIMA model tracking the regular weekday/weekend rhythm. Several sharp error spikes of MAE and RMSE coincide with high volatile periods of 2024 as the dataset analysis indicated, resulting in poor performance on those months, with the model unable to fully adapt on those changes.

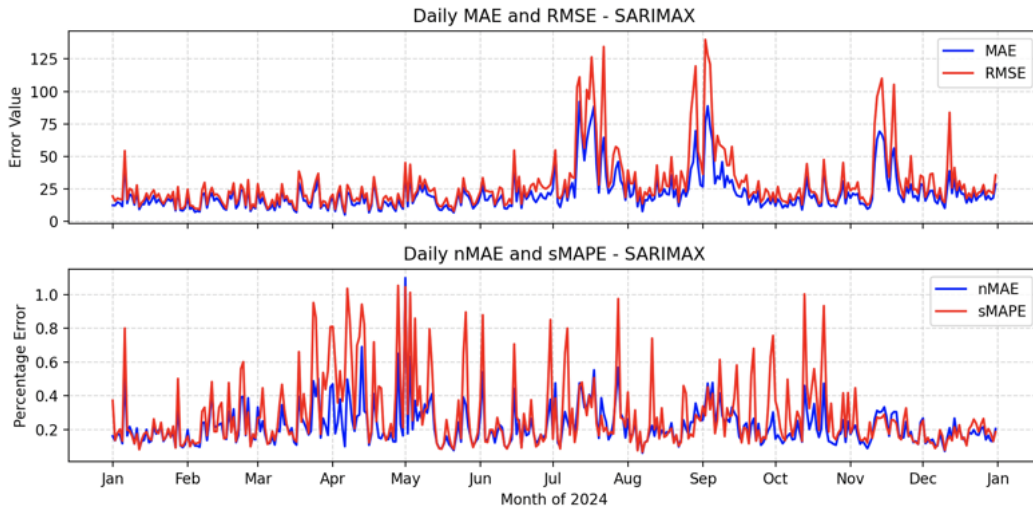


Figure 14: Evaluation Metrics Daily average - SARIMAX.

6.1.3 Random Forest Results

The results of the Random Forest method for the different predictor sets are found on Table 4. Univariate Random Forest already cuts the benchmark MAE by 32%, demonstrating the ensemble’s ability to distill complex autoregressive structure from raw price lags alone. The massive improvement in accuracy translates also to the rest of the evaluation metrics. Adding the day-ahead forecasts of load, and wind/solar generation shaves further $\sim 3\%$ off MAE and $\sim 2\%$ off RMSE. Further incorporating the calendar dummies on top of the previous predictors lowers MAE

to 13.85 €/MWh and an nMAE of 0.137 making this the best of the three Random Forest variants proving the importance of the selected features. Overall, the fully featured Random Forest improves the Persistence method by 36% in MAE and 27% in RMSE.

Model	MAE	RMSE	sMAPE	nMAE	Time
Persistence	21.860	40.860	27.423	0.217	–
Random Forest (univariate)	14.849	31.286	20.112	0.147	~ 22 min
Random Forest (Load, Solar, Wind)	14.428	30.742	19.627	0.143	~ 34 min
Random Forest (all predictors)	13.851	29.691	18.939	0.137	~ 38 min

Table 4: Performance metrics for Random Forest models.

The monthly timeseries of the metrics in Figure 15 show the average daily price of MAE and RMSE hover around 10 €/MWh with those errors increasing on months July, September and November where the most extreme prices of 2024 were recorded as per the dataset analysis.

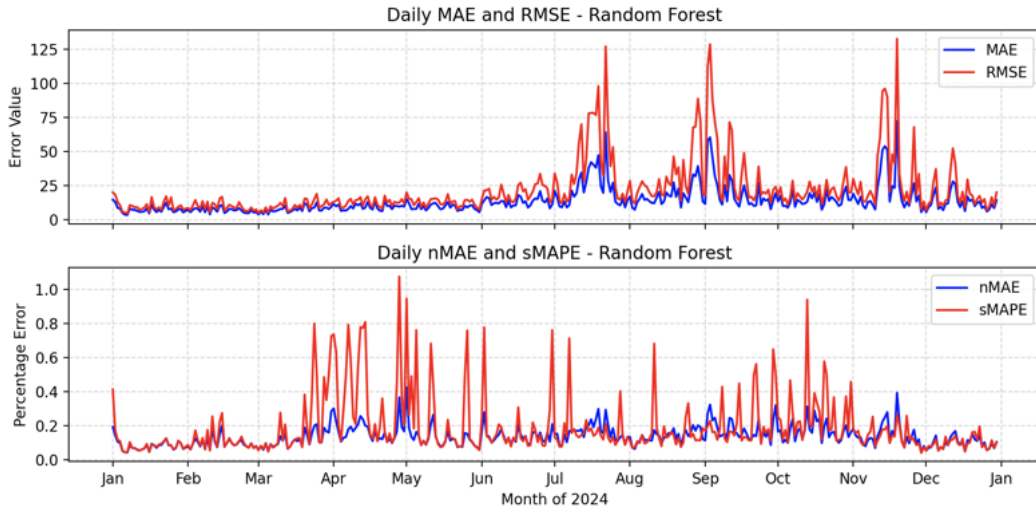


Figure 15: Evaluation Metrics Daily average - Random Forest.

6.1.4 XGBoost Results

The results of the XGBoost method for the different predictor sets are found on Table 5. Univariate XGBoost already cuts the benchmark MAE by 30%, demonstrating the ensemble’s ability to distill complex autoregressive structure from raw price lags alone. These massive improvement in accuracy translates also to the rest of the evaluation metrics. Adding the day-ahead forecasts of load, and wind/solar generation reduces further MAE $\sim 7\%$ and RMSE $\sim 8\%$. Further incorporating the calendar dummies on top of the previous predictors lowers MAE to 12.50 €/MWh and an nMAE of 0.124 making this the best of the three XGBoost variants and proving the importance of the selected features. Overall, the fully featured XGBoost improves the Persistence method by 43% in MAE and 33% in RMSE and is the “champion” configuration selected for this method.

Model	MAE	RMSE	sMAPE	nMAE	Time
Persistence	21.860	40.860	27.423	0.217	–
XGBoost (univariate)	15.225	31.801	20.416	0.151	~ 26 min
XGBoost (Load, Solar, Wind)	14.174	29.391	19.803	0.141	~ 36 min
XGBoost (all predictors)	12.502	27.186	17.662	0.124	~ 39 min

Table 5: Performance metrics for XGBoost models.

The monthly timeseries of the metrics in Figure 16 show the average daily price of MAE and RMSE hover around 8 €/MWh with those errors increasing on months July, September and November where the most extreme prices of 2024 were recorded as per the dataset analysis.

6.1.5 LSTM Results

The results of the LSTM neural networks for the different predictor sets are found on Table 6. Purely univariate LSTM performs worse than the naive benchmark on every metric. The network does learn the weekly rhythm but appears to over-smooth sharp peak, yielding larger MAE and nMAE. Adding the day-ahead forecasts of load, and wind/solar generation reduces both MAE by $\sim 3\%$ and RMSE by $\sim 2.8\%$, yet the

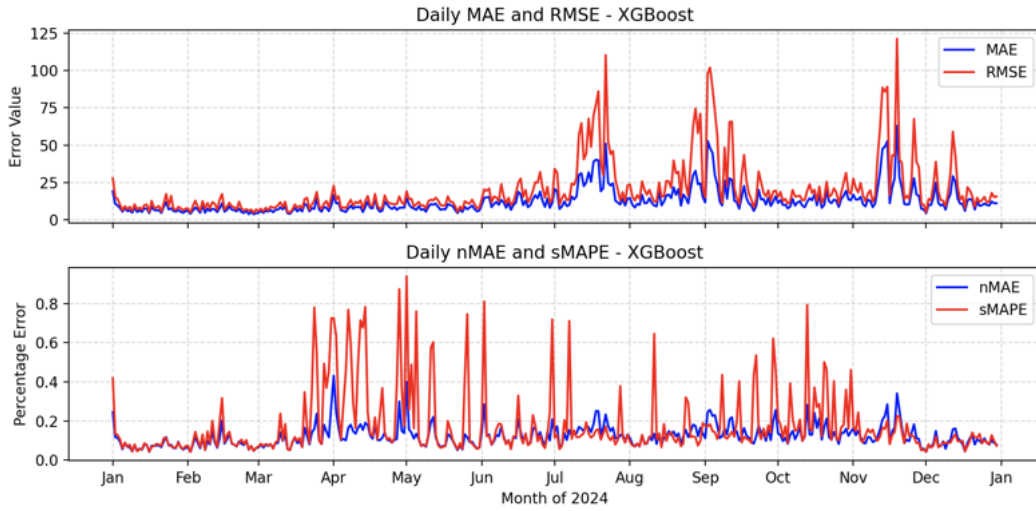


Figure 16: Evaluation Metrics Daily average - XGBoost.

model still trails the benchmark. Finally, after incorporating the calendar dummies on top of the rest predictors into the feature tensor finally nudges the LSTM slightly ahead of the Persistence method. MAE drops to 20.96 €/MWh and RMSE to 38.455 €/MWh. This is also selected as the “champion” configuration and although it improves the accuracy, the results are comparative to the ones obtained from the benchmark.

Model	MAE	RMSE	sMAPE	nMAE	Time
Persistence	21.860	40.860	27.423	0.217	–
LSTM (univariate)	23.209	40.211	28.245	0.230	~ 17 min
LSTM (Load, Solar, Wind)	22.478	39.096	28.216	0.223	~ 19 min
LSTM (all predictors)	20.958	38.455	26.587	0.208	~ 21 min

Table 6: Performance metrics for LSTM models.

The daily timeseries of the metrics in Figure 17 show the average daily price of MAE and RMSE hover around 10-20 €/MWh with those errors increasing on months July, September and November where the most extreme prices of 2024 were recorded as per the dataset analysis

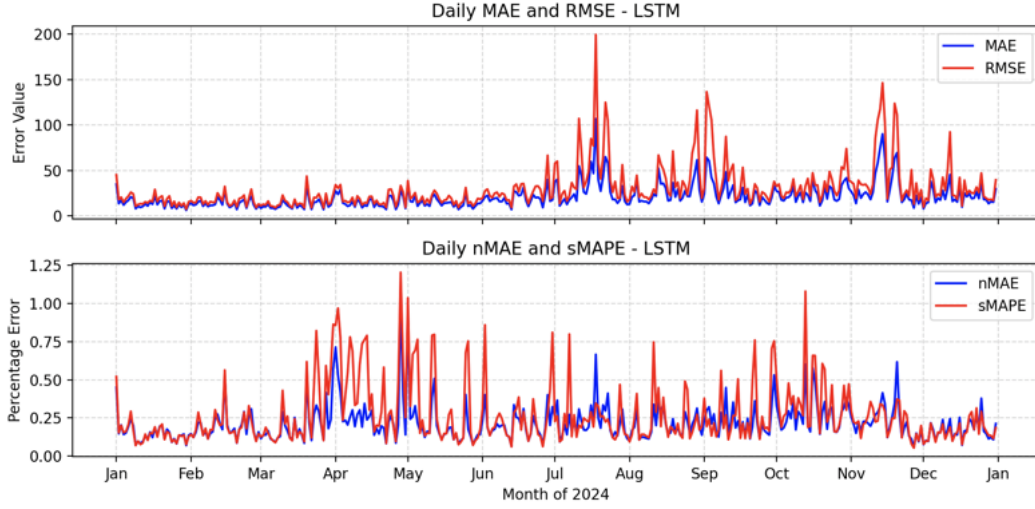


Figure 17: Evaluation Metrics Daily average - LSTM.

6.2 Model Comparison and Performance Overview

This section contrasts the “champion” configurations of every modeling family introduced in Section 6.1 against the common 2024 test horizon. The comparison begins in Table 7 with the global accuracy metrics being presented and immediately indicating a clear winner in all performance metrics. XGBoost establishes the accuracy frontier with a 43% reduction in MAE and a 33% drop in RMSE relative to the Persistence method, while keeping computation time under forty minutes. Random Forest follows closely, delivering a 37% MAE improvement, with slightly higher tail RMSE. SARIMAX provides only cosmetic gains over the benchmark, confirming that linear seasonality plus extra predictors cannot fully exploit the complex drivers of day-ahead prices. Finally, LSTM edges past Persistence by ~ 1 €/MWh but lags well behind the tree ensembles despite GPU acceleration, indicating that the network’s expressive power did not translate into superior generalization for this task.

To support further that matchup, the comparison moves to seasonal behavior by calendar month in Table 8 and finally intra-week patterns by day-of-week in Table 9. XGBoost is yet again best-in-class for every month with its advantage peak in winter but also remain sizable during the volatile summer. Random Forest is the consistent runner-up, usually within 1-2 €/MWh of XGBoost in the low-volatility months yet falling further behind during July-September. Summer remains challenging for

Model	MAE	RMSE	sMAPE	nMAE	Time
Persistence	21.860	40.860	27.423	0.217	–
SARIMAX	21.670	36.681	28.982	0.215	~ 38 min
Random Forest	13.851	29.691	18.939	0.137	~ 38 min
XGBoost	12.502	27.186	17.662	0.124	~ 39 min
LSTM	20.958	38.455	26.587	0.208	~ 21 min

Table 7: Performance metrics for best models.

all models as July MAE balloons across the board, but the tree ensembles achieve roughly half the Persistence error. SARIMAX and LSTM struggle disproportionately in extreme months. In August, for example, both record MAE above 25 €/MWh with the Persistence being at 23.61€/MWh.

Month	Persistence	SARIMAX	Random Forest	XGBoost	LSTM
January	14.22	15.78	7.60	7.20	13.74
February	11.41	15.06	7.12	6.28	13.77
March	11.74	15.58	7.65	6.98	12.50
April	15.45	14.56	9.63	8.62	15.20
May	15.69	17.56	9.64	8.07	14.74
June	17.27	18.72	13.35	12.10	19.30
July	41.16	40.35	22.77	21.07	32.22
August	23.61	25.13	18.01	15.60	25.90
September	28.64	29.88	20.56	18.55	25.56
October	21.92	17.33	14.60	12.29	24.44
November	36.57	29.20	21.13	20.81	32.40
December	24.31	20.65	12.63	12.38	21.55

Table 8: Monthly MAE values for the different forecasting models.

The key observations from the day-wise MAE table are that Monday and Tuesday are the hardest to predict, as all models incur their largest weekday errors. XGBoost again dominates every day, with the margin widest on weekends and Random Forest mirrors the pattern but at a slightly higher error level. LSTM performance is noticeably improved in comparison to the Persistence and SARIMAX on the weekends. Although all the evaluation metrics picked in Section 5.2 were tracked, we concentrate on MAE as there is a common scale across models, meaning that all forecasts are evaluated on the same 2024 price series, so MAE expresses the average miss directly in €/MWh, a unit that can be interpreted without additional normalization.

Day	Persistence	SARIMAX	Random Forest	XGBoost	LSTM
Monday	25.02	24.22	16.91	14.70	24.07
Tuesday	22.65	23.12	15.55	13.93	22.91
Wednesday	22.21	21.09	15.15	13.70	22.79
Thursday	21.20	23.20	14.30	13.12	23.39
Friday	20.05	19.88	13.08	12.23	18.78
Saturday	20.50	20.10	10.14	9.40	16.25
Sunday	21.30	20.00	10.96	10.39	18.43

Table 9: Daily MAE values for the different forecasting models.

To further enhance on the comparison of hourly MAE distribution for a method by month we present Figure 18 with each panel including the box plots of MAE for each month of 2024 (boxes = inter-quartile range, whiskers = 1.5xIQR, black line = median, red line = monthly mean). The key take always from these plots are that during January and April all models perform best with XGBoost and Random Forest reducing hourly errors to single-digits for most of the period. For SARIMAX, medians dip 1-2 €/MWh below the benchmark in winter but the IQR and tails remain almost identical, especially during the July spike. Random Forest compresses the entire error distribution with winter boxes shrink to 2-6 €/MWh and even at the July peak the upper whisker stops below 50 €/MWh, roughly half the benchmark's extreme. XGBoost is consistently the tightest model, with IQRs stay under 8€/MWh for nine months of the year, medians fall below 5€/MWh from January through May. For LSTM January - March medians are ~ 10 €/MW, slightly better than the Persistence, yet boxes and whiskers become progressively taller in June - November.

The seasonal boxplots we analyzed hide observations beyond 1.5xIQR so the bulk of the distribution remains legible. To understand where those hidden points occur, we isolated the top 1% actual prices in the 2024 test set (threshold = 382€/MWh) and all forecast outliers whose MAE exceeded 150€/MWh. The two resulting histograms are displayed in Figure 19. Observing these figures shows that same clock-hours dominate both plots. More than 90% of the record-high prices occur between 19:00 and 22:00, with a secondary dip at 8:00-9:00 and an isolated cluster around 17:00-18:00. All five models register their largest MAE values at exactly those hours. XGBoost and Random Forest still produce extreme misses in the evening peak, but they do so 25-30% less often than SARIMAX and Persistence method, with LSTM sitting in between. In short, the alignment between price extremes and error

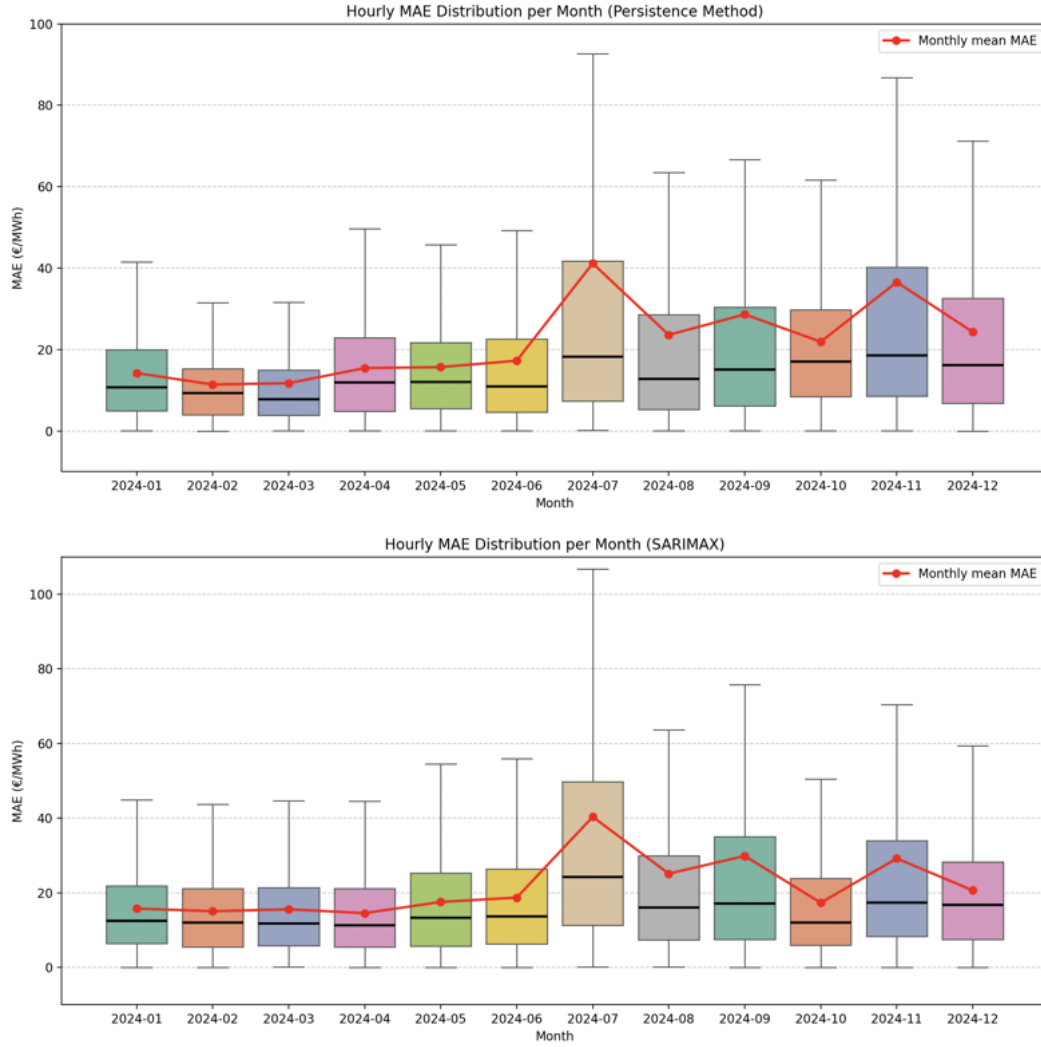


Figure 18: Monthly MAE boxplots for all models.

extremes confirms that our models are chiefly limited by the unpredictability of rare but powerful market shocks, and not by their fundamental capacity to learn and extract regular patterns in the data.

To move from summary statistics to operational insight we zoom in on the two most turbulent episodes of 2024, early July where demand pushed the DAM above 550 €/MWh and the first week of September where the price spikes exceeded 900 €/MWh. Figure 20 plots the actual hourly DAM price against the forecasts of each champion model for those two periods. By observing these results, we see that Persistence collapses when a new regime emerges and because it merely repeats the previous week's level it under-forecasts the 12 July super-peak by >300€/MWh.

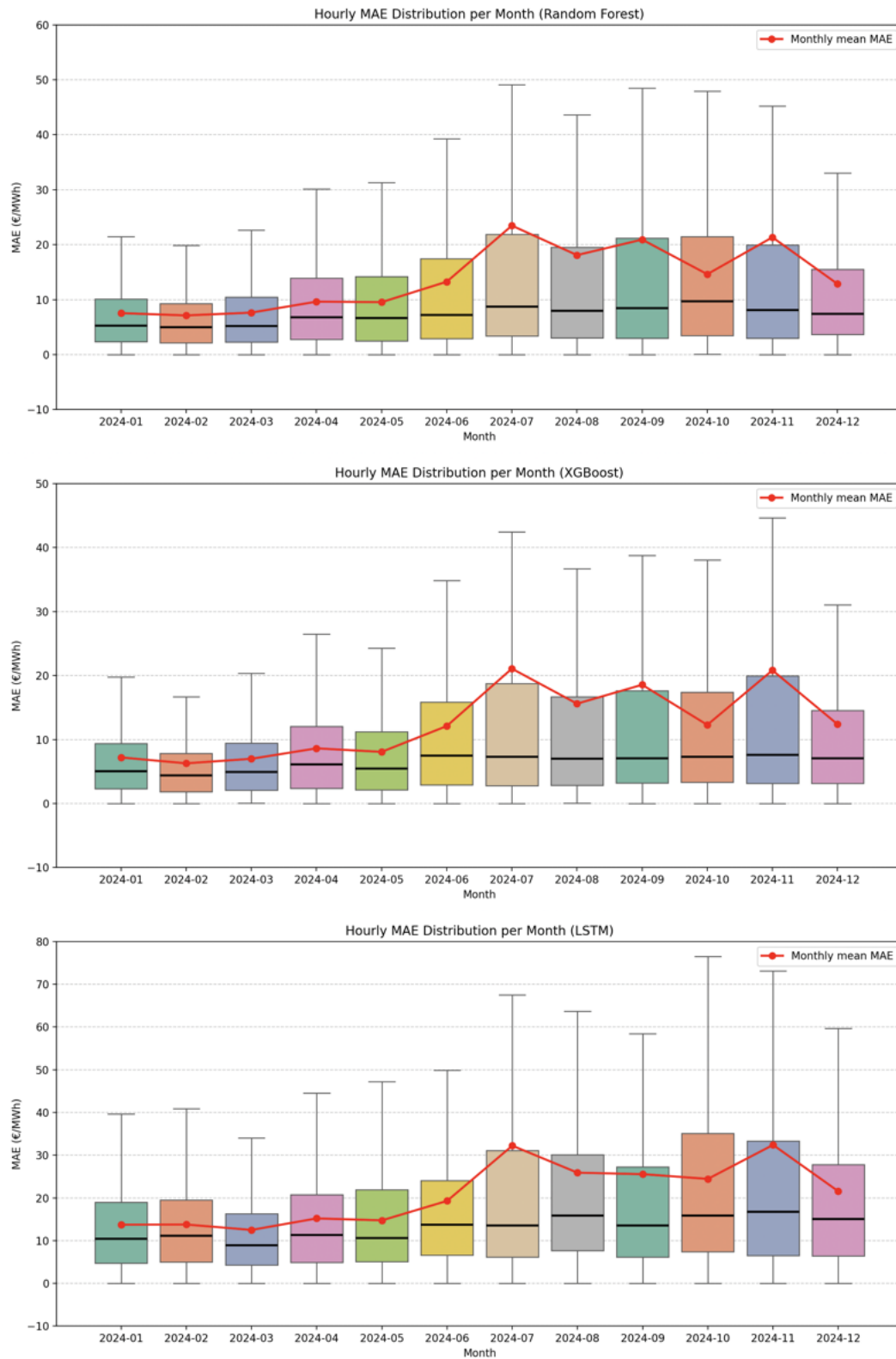


Figure 18: Monthly MAE boxplots for all models (continued).

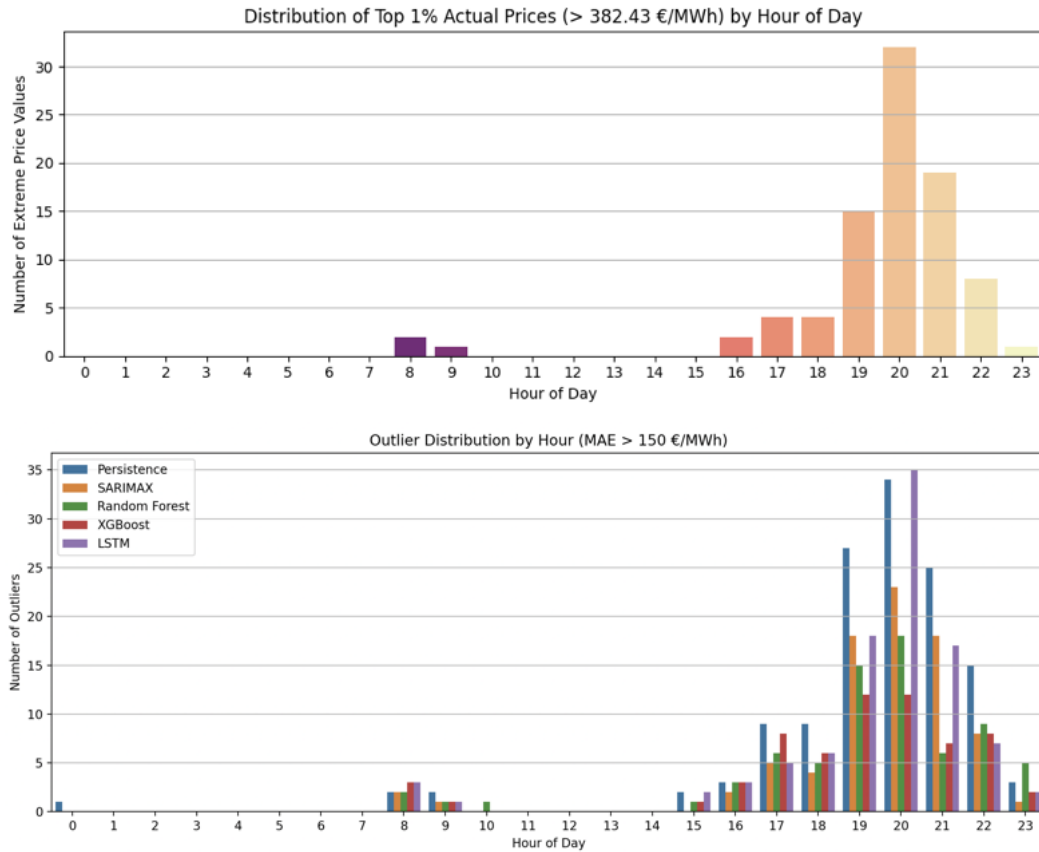


Figure 19: Top actual prices of 2024 distribution (Top) and extreme MAE outliers distribution (Bottom).

SARIMAX seems able to adapt to shape, but not amplitude with post-spike decay being too slow. Random Forest tracks peaks better and it captures the sequence of consecutive spikes >500 €/MWh better as well. XGBoost is consistently closest and it achieves the best amplitude among all models and on 09/05 it predicts 860 €/MWh vs an actual 910 €/MWh.

In general, all supervised models anticipate which hours will experience turmoil, but the difference lies in how accurately they gauge the amplitude. For example, SARIMAX times the event but lacks the non-linear head room to scale beyond historical maxima. LSTM's alternating over/under shoot suggests it learns local momentum but cannot distinguish temporary scarcity from new normal. Non of the models fully captures the 900 €/MWh September crest and incorporating extra features like cross-border flows or intra-day variability may be necessary to bridge the last accuracy gap as when the market enters unprecedented territory, model flexibility matters.

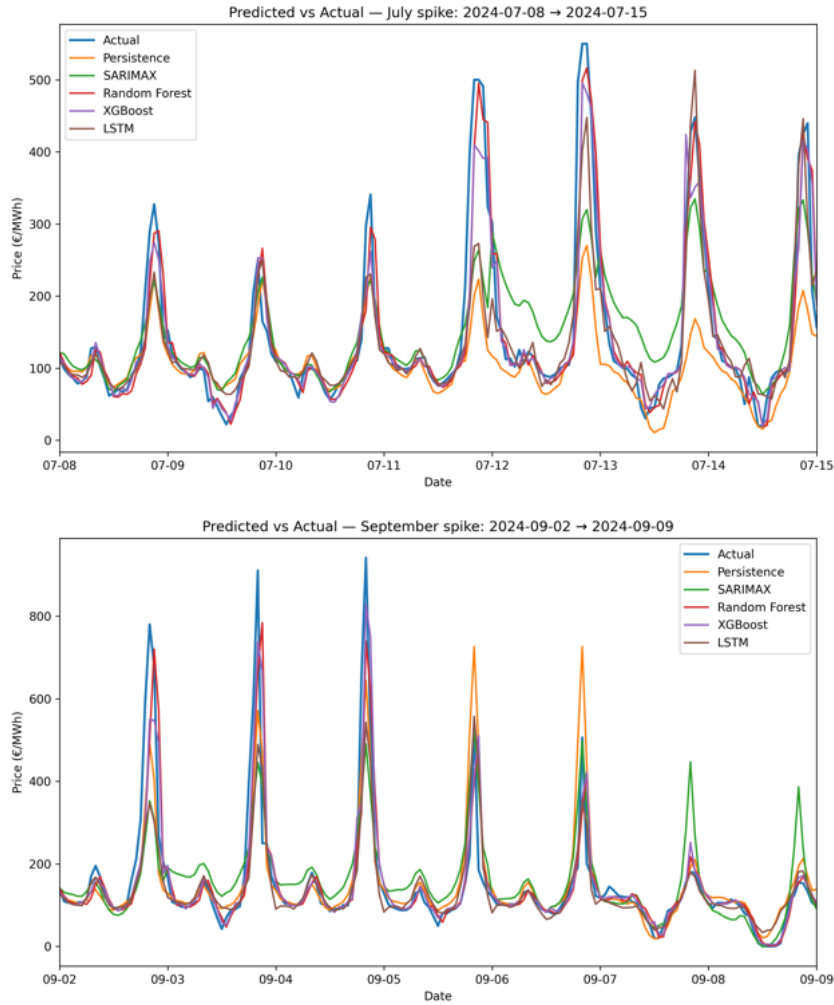


Figure 20: 2024 extreme spike periods, July (Top) and September (Bottom).

Descriptive statistics like MAE or RMSE show how much one model beats another, but they do not tell us whether the observed gap could have risen by chance. The Diebold-Mariano (DM) test answers that question by comparing two competing forecasts period-by-period over the same evaluation window, forming the series of loss differentials of the error metric (here MAE) and adjusting for any autocorrelation that arises in multi-step horizons (here 24h). The returning p-value tests the null hypothesis H_0 that the two competing forecasts have equal predictive power. A small p-value (<0.05) allows to reject the H_0 and conclude that the two models are statistically different. The results of the DM test of our models in found on Figure 21.

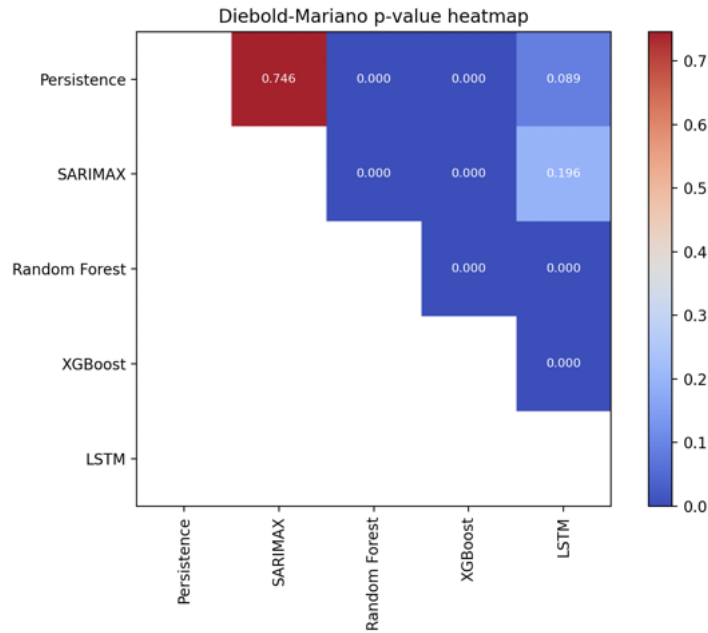


Figure 21: Diebold Mariano test.

Persistence vs SARIMAX fail to reject the null hypothesis meaning that both classical benchmarks perform similarly at the 24 h horizon. Regarding Persistence vs LSTM there is weak evidence that LSTM beats Persistence, which is not enough at the conventional 5% threshold. Finally all tree ensembles decisively outperform all other models with the gap between XGBoost and Random Forest being real, as the test confirms, with XGBoost beating Random Forest.

To gauge each model's dependence on sample size, each "champion" configuration was re-estimated on five progressively shorter windows, from the full 2019-2023 history down to just the year 2023, while holding the 2024 test set fixed. Figure 22 plots the resulting MAE curves. XGBoost and Random Forest curves are almost flat as MAE drifts by <0.2 €/MWh across all window lengths. The conclusion is that tree ensembles extract most of their signal from recent patterns and calendar/forecast features. Once a couple of annual cycles are seen, adding older regimes yields diminishing returns. SARIMAX worsens as we drop pre-2022 data and then improves when only 2023 remains, indicating that the energy-crisis of 2022 affects the performance of the model. The most important observation about this analysis is regarding the behavior of LSTM, as it collapses when the window contracts to one year. Neural networks are data-hungry and are affected by parameters to samples ratio. With so little variety the network over-fits daily noise, forgets rarely occurring

holiday and spike patterns and generalizes poorly on 2024 test set. Longer histories supplies the network with multiple instances of each intra-day shape and enough extreme events to calibrate high price tail.

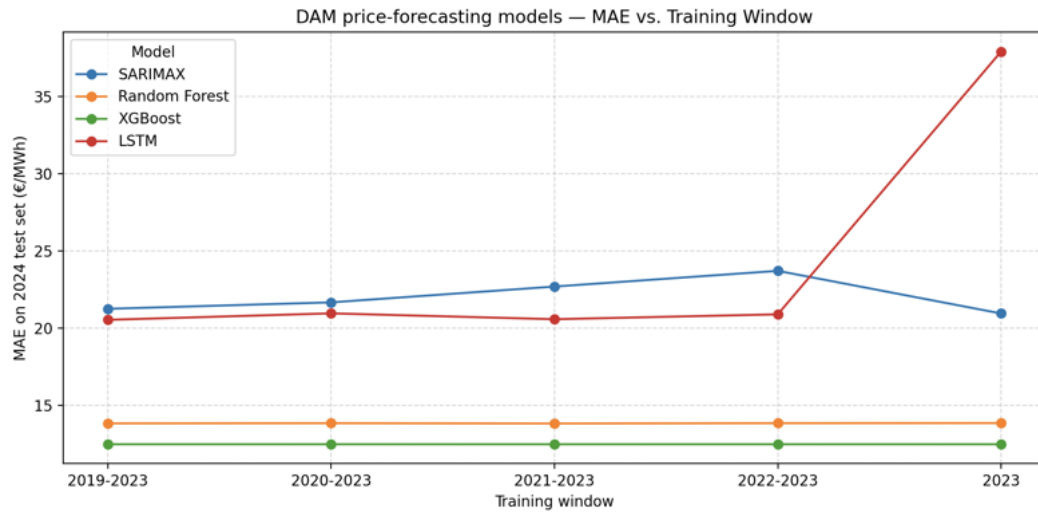


Figure 22: Average MAE on different training windows.

7 Conclusions and Future Work

7.1 Conclusions of this Thesis

This thesis set out to compare four widely used forecasting paradigms, SARIMAX, Random Forest, XGBoost and LSTM on the task of day-ahead price forecasting for the Greek wholesale electricity market over the period 20220-2024. From the extensive experimental campaign and the statistical tests that followed, several firm conclusions can be drawn.

Tree-based ensemble methods are clearly superior for the Greek DAM. XGBoost delivered the lowest overall error outperforming the naive weekly-persistence benchmark by 43% in MAE and 33% in RMSE. Random Forest followed closely, confirming the robustness of ensemble decision trees.

Linear seasonal order models add limited incremental value once non-linear alternatives are available. A careful tuned SARIMAX with Fourier terms and exogenous Load/RES forecasts narrowed the gap to the benchmark yet remained $\sim 30\%$ less accurate than XGBoost. The method remains attractive for its transparency and very modest computational cost, but its linear structure struggles with the highly non-stationary, spike prone price series that characterized 2022-2024.

Deep recurrent neural networks under-performed in this setting. A single-layer LSTM with 64 cells and 168-h look-back window did not beat the naïve baseline until calendar dummies and exogenous drivers were injected, and even then, marginally. Data hunger, sensitivity to hyper-parameters and relative scarcity of extreme events in the training window appeared to have hampered generalization of this method.

Adding exogenous, physically meaning predictors proved to be crucial. Across all model classes, the inclusion of day-ahead system-load forecast and technology specific RES forecasts consistently reduced both average and tail errors. Calendar descriptors further tightened residuals by helping the algorithms anchor recurring behavioral patterns.

The results obtained in this thesis confirm the strong forecasting ability of tree-based

methods, with XGBoost emerging as the most effective model. Nevertheless, it is difficult and probably not entirely accurate to directly compare our results with those reported by other researchers in the literature review, since each study investigates different methods under distinct market dynamics, time periods, and price scales. Despite these differences, we attempted a relative comparison by examining how the best methods in the respective studies improved their own Persistence (or naive) benchmarks.

In [28], the best method achieved a reduction of the MAE from 12.240 to 5.639 and the RMSE from 15.156 to 10.401, corresponding to improvements of 53.94% and 31.37% respectively. In [21], the LSTM model reduced the RMSE from 31.6256 to 7.2073, yielding an improvement of 77.2%. In [24] the GRU model lowered MAE from 7.95 to 5.71, corresponding to a 28.1% improvement. In this thesis, as stated earlier, the XGBoost model achieved a reduction of the MAE by 43% and of the RMSE by 33%, compared to the Persistence benchmark.

Although the absolute values differ because of the heterogeneity of the datasets and market conditions, the relative improvements are of similar magnitude. This shows that the performance gains of XGBoost in the Greek Day-Ahead Market are broadly consistent with those reported in the literature, reinforcing the conclusion that advanced machine learning method, particularly tree-based ensembles, represent a robust and effective approach to electricity price forecasting. Ideally, such comparisons would rely on normalized error metrics to ensure comparability across studies, but lack of consistency in the literature regarding the evaluation criteria makes this approach unfeasible in the present case.

7.2 Discussion and Future Work

While the study provides a clear performance ranking for today's Greek DAM, it motivates the further examination of the electricity price forecasting endeavor under Greek market dynamics. Experimenting and incorporating extra exogenous data like fuel-price futures, cross-border capacity allocation, balancing market prices and intraday flow nominations could improve the accuracy of the models and could be the obvious next step.

Hybrid and attention-based deep architectures should also be one of the next steps. Convolutional-LSTM hybrids and Temporal Fusion Transformers (TFT) have recently broken performance records on other power system datasets by automatically learning variable importance and long-range dependencies. A GPU -accelerated benchmark of these models using the same engineered features plus the ones suggested above, would clarify whether deep learning can finally surpass boosted trees once data richness is sufficient.

Moreover, Day-ahead actors are risk-averse, and single-point forecasts do not expose the uncertainty that surrounds extreme spikes. Extending the XGBoost framework to quantile regression or employing gradient-boosted NGBoost and LightGBM-QR variants would yield full predictive distributions, supporting value-at-risk, reserve sizing and bidding-strategy optimization.

Finally high temporal resolution is a must as the forthcoming shift of the European SDAC to 15-min granularity will expose additional intra-hour volatility and feature lags and evaluation methodology must be adapted. Investigating hierarchical models that coherently forecast 15-min, hourly and daily aggregates would add value for portfolio optimization.

The empirical evidence in this thesis provides a solid springboard, but the rapidly changing energy landscape ensures that electricity-price forecasting will remain an evolving and fertile research field.

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